Bridge-Valence Diagnostics: A Relation-First Method for Classifying Bridge Problems

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Abstract

The paper *Critical Relation Loci and Bridge-Valence in Observer Slices* introduced the idea that persistent residualization can indicate a critical relation locus. The main risk is that this could be dismissed as a convenient classification invented after encountering a hard problem.

This paper strengthens the idea into a replicable diagnostic.

The diagnostic has two stages.

First, a **pre-solution target-demand diagnostic** predicts whether a problem is likely to be a bounded bridge, a finite unification atlas, or a critical relation locus before the bridge is solved. It uses only declared target structure:

$$\boxed{\mathcal{D}_{\text{pre}}(X, D) = (\rho_{\text{face}}, b_{\text{targ}}, s_{\text{sub}}, n_{\text{NFT}}, \kappa_{\text{compat}}, h_{\text{obs}}).}$$

Second, a **post-diagnostic residual persistence diagnostic** confirms whether the prediction is structural:

$$\boxed{\mathcal{D}_{\text{post}} = (r_{\text{persist}}, g_{\text{bridge}}, a_{\text{diag}}).}$$

A problem is predicted to be a critical locus when it demands multiple independent target faces, activates no-free-transfer across several target types, lacks a single declared compatibility operator collapsing those target types, and has high observer-slice bridge-valence.

The paper then applies the diagnostic schematically to several known cases:

1. Newtonian mechanics as a bounded bridge inside its declared target regime;
2. Maxwellian electromagnetism as a finite unification atlas: crossroads-structured but stabilized by field equations and common compatibility laws;
3. Standard Model gauge structure as a high-valence finite unification atlas stabilized by gauge symmetry and representation structure but with residuals outside the declared target;
4. the observer-slice Alexandrov/continuum locus as a critical relation locus whose proper closure is a branching target atlas.

The conclusion is:

$$\boxed{\text{criticality is not assigned because a problem is hard;}}$$

it is predicted when the target-demand matrix has high bridge-valence and no terminal compatibility operator for all demanded target faces.

The diagnostic is explicitly partial and error-bounded. It returns an admitted classification only when declaration completeness, diagnostic margin, and residual behavior are adequate for the claimed diagnostic status. The diagnostic output is therefore:

$$\boxed{\text{Diag}(X, D) = (C, \Delta_{\text{diag}}, m_C, \text{Status}).}$$

Here C is a candidate class, Δ_{diag} is the diagnostic-error vector, m_C is the classification margin, and Status records whether the classification is admitted, weak, blocked, or unresolved.

The six pre-diagnostic coordinates are derived from existing programme machinery rather than chosen freely:

$$\rho_{\text{face}} \leftarrow \text{triadic face-demand / NUR–NUB–NUO;}$$

$$b_{\text{targ}} \leftarrow \text{bridge accessibility and target-presentation valence;}$$

$$s_{\text{sub}} \leftarrow \text{triadic substitution / role-recursion depth;}$$

$$n_{\text{NFT}} \leftarrow \text{No-Free-Transfer and structure-type no-promotion;}$$

$$\kappa_{\text{compat}} \leftarrow \text{valued face-recovery / compatibility-operator machinery;}$$

$$h_{\text{obs}} \leftarrow \text{observer/readout, chart, evaluator, and calibrator burden.}$$

Thus the diagnostic is not an imposed scoring rubric. It is the readout vector naturally forced when a relation-first programme audits a bridge claim.

The diagnostic also includes recursive theorem/no-go activation. A diagnostic state is:

$$\boxed{\mathfrak{D}_n = (\mathcal{D}_{\text{pre}}^{(n)}, \mathcal{D}_{\text{post}}^{(n)}, \Delta_{\text{diag}}^{(n)}, \mathcal{A}_n, \mathcal{B}_n, \mathsf{Status}_n),}$$

with update:

$$\boxed{\mathfrak{D}_{n+1} = U_{\text{diag}}(\mathfrak{D}_n, \Omega_n, \mathcal{T}_{\text{prog}}).}$$

Here $\mathcal{T}_{\text{prog}}$ is the programme theorem/no-go tree, $\mathcal{A}_n$ is the active theorem/no-go subtree, and $\mathcal{B}_n$ is the current branch field. The recursive process stops only at declared statuses such as

$$\boxed{\text{closed}, \quad \text{blocked}, \quad \text{bounded/residualized}, \quad \text{persistent branching.}}$$

The landscape table is not a historical derivation. It is a diagnostic calibration atlas: a compact reconstruction table for asking which target faces, residual types, theorem/no-go activations, and status margins are active in each case.

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0. Source basis

Source	SHA256 prefix	Status
critical_relation_loci_bridge_valence_observer_atlas.md	627a487a0298e882	source basis
bridge_accessibility_target_relative_inaccessibility_bridge_reader_facing.md	81191fd275b26298	source basis
programme_wide_result_atlas_reciprocal_shape_missing_face_pass.md	6acdf70c01f4971a	loaded
programme_wide_result_atlas_reciprocal_shape_missing_face_pass2.md	e2b5ca0932d09f19	loaded
programme_wide_result_atlas_reciprocal_shape_missing_face_pass3.md	815e6a3239b70492	loaded

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This source basis lists programme papers and prior mathematical artifacts, not internal workflow notes.

0B. Canonical programme dependency spine

The source basis above records local artifacts used in the present draft. The mathematical dependency spine is the following programme paper stack.

Programme paper / theorem family	Role in this paper
Theory of Organization	relation-first organization, proper non-boundary layer, task-relative organization
Charted Organization Theory	charts as relation-maps, observer as admissible positioned chart, task-visible delineation, chart/atlas readout
Recursive Chart Learning Theory	trace-admitted branches, evaluator/admission/update modes, recursion as path-trace-induced change
Recursive Bridge Calibration	source–bridge–target decomposition, calibrator, target distinction family, admission/update, obstruction vector, bridge-selection discipline
Triadic Closure / Recursive Residual Theory	relation-to-triad declaration, primitive $(\mathsf{Z}, \mathsf{R}, \mathsf{O})$, closure-incomplete subfamilies, residual recursion pressure
Triadic Substitution, Coupling, and Closure Atlases	role-substitution, typed micro-triads, landed coupling, closure atlas, no dangling role interface
No Undeclared Relation	no bridge without declared mediation, boundary, observer/readout, and target distinction
No Undeclared Boundary	no exactness, approximation, or admissibility without declared boundary/tolerance/support
No Undeclared Observer	no visibility, readout, evaluator, chart, or calibration without declared observer/readout role
No-Free-Transfer / structure-type no-promotion	no target-type transfer without a declared bridge law
Bridge Accessibility and Target-Relative Inaccessibility	bridgedness as admitted target accessibility; unbridgedness as target-relative inaccessibility
Critical Relation Loci and Bridge-Valence	persistent residualization, bridge-valence, critical loci, nonterminal branching target atlases
Target-Presentation Calibration Atlas	comparison of observer-slice AC presentation to multiple target presentations and residual families

The present paper is not a replacement for these papers. It is a diagnostic layer that reads their machinery as a repeatable classification procedure for bridge problems.

0C. Cohesion of the paper

The paper has grown through several patches, so its internal dependency must be explicit.

0C.1 Central dependency chain

The central dependency chain is:

$$\boxed{\text{relation-first} \Rightarrow (\mathsf{Z}, \mathsf{R}, \mathsf{O}) \Rightarrow \text{charted readout} \Rightarrow \text{recursive branch evaluation} \Rightarrow \text{bridge calibration} \Rightarrow \text{diagnostic}}$$

Every later component is a readout, recursion, or calibration of the earlier components.

0C.2 Component map

Paper component	Depends on	Output
target-demand matrix	relation-to-triad + target distinctions	active face/target demands
diagnostic coordinates	target-demand matrix + no-free-transfer + observer/readout	$(\rho, b, s, n, \kappa, h)$
recursive diagnostic	RCLT + RBC + theorem/no-go tree	diagnostic update $\mathfrak{D}_{n+1} = U_{\text{diag}}(\mathfrak{D}_n, \Omega_n, \mathcal{T}_{\text{prog}})$
diagnostic error/status	calibrator + boundary/readout discipline	partial/error-bounded classification
diagnostic table	coordinates + target regimes + theorem/no-go activation	reproducible landscape of cases
scoring calibrator	Charted Organization + RCLT + RBC	$\mathfrak{K}^{\Gamma}_{\text{diag}}$
recursive triadic topology	triadic substitution + closure atlas	topology read by score charts
fractal guardrail	no undeclared scale/readout + calibrator discipline	no literal fractal claim without scale/self-similarity data

Theorem 0C.3 — Cohesion Theorem

Every formal component of the diagnostic paper is obtained by one of four programme-native operations:

1. triadic declaration;
2. charted readout;
3. recursive trace/update;
4. calibrated admission.

Therefore the paper is a cohesive diagnostic layer over the existing programme machinery rather than a collection of independent heuristics.

Proof

The target-demand matrix is triadic declaration of target requirements. The diagnostic coordinates are charted readouts of those requirements. The recursive diagnostic state and update are trace/update operations inherited from Recursive Chart Learning and Recursive Bridge Calibration. Diagnostic error, admission, and scoring are calibrator operations. The table is an application of the diagnostic calibrator to declared target regimes. The recursive triadic topology is the closure-atlas presentation of repeated triadic substitution and coupling. Thus every component is generated by triadic declaration, charted readout, recursive update, or calibrated admission. □

Corollary 0C.4 — No independent heuristic layer

The diagnostic may use declared scoring charts, but those scoring charts are calibrator/readout choices over programme-derived coordinates. They do not constitute an independent heuristic layer.

Corollary 0C.5 — Revision without incoherence

Future revisions may refine the scoring granularity, table rows, or target regimes without breaking the diagnostic, provided the refinement is a declared chart/calibrator update over the same programme-derived coordinates.

0A. What “diagnostic audit” means in this paper

This paper uses the phrase **diagnostic audit** in a technical sense.

It does **not** mean the internal paper-construction workflow audit. It means the mathematical diagnostic procedure defined in this paper:

$$\boxed{\text{diagnostic audit} = \text{relation-first check of a target-bridge problem by the programme theorem/no-go tree.}}$$

A diagnostic audit has five steps:

1. declare the source presentation X;
2. declare the target regime D'_Y;
3. compute the diagnostic coordinates:

$$(\rho_{\text{face}}, b_{\text{targ}}, s_{\text{sub}}, n_{\text{NFT}}, \kappa_{\text{compat}}, h_{\text{obs}});$$

4. activate the relevant theorem/no-go subtrees;
5. return an admitted, conditional, borderline, mixed, blocked, diagnostic-pending, or insufficiently declared status.

Thus, when this paper says a residual “survives diagnostic audit,” it means:

$$\boxed{\text{it survives the relation-first diagnostic procedure above.}}$$

It does not mean that an internal workflow document is a mathematical source.

0A.1 Programme-paper dependency map

The diagnostic coordinates come from programme papers/theorem families as follows:

Diagnostic part	Programme source machinery
ρ_{face}	relation-to-triad declaration; NUR/NUB/NUO; closure-incomplete subconfiguration no-gos
b_{targ}	bridge accessibility; target-presentation calibration; reciprocal mediation relation-class
s_{sub}	triadic substitution; role recursion; no dangling role interface
n_{NFT}	No-Free-Transfer; structure-type no-promotion; quotient admissibility
κ_{compat}	valued face-recovery; compatibility operators; residual/quotient algebra
h_{obs}	Charted Organization; observer/readout; evaluator/calibrator; empirical-readout residuals
recursive update U_{diag}	Recursive Chart Learning; Recursive Bridge Calibration; closure-atlas branch updates
diagnostic error Δ_{diag}	boundary/admission discipline; calibrator residuals; diagnostic status theorem

This is the mathematical source basis for the diagnostic.

1. Problem

The previous critical-locus paper introduced:

$$\boxed{\text{persistent residualization} \Rightarrow \text{candidate critical relation locus} \Rightarrow \text{branching target atlas.}}$$

The objection is serious:

$$\boxed{\text{How do we know this is not a convenient label applied after failure?}}$$

The answer must be diagnostic and predictive. A researcher using the programme should be able to inspect a proposed bridge and predict whether it is likely to be:

1. a bounded bridge;
2. a finite unification atlas;
3. a critical relation locus;
4. or an accidental underdeclaration.

This paper defines such a diagnostic.

1A. Programme-native derivation of the diagnostic

The diagnostic must not be an external classifier imposed on the programme. It must be derived from the programme's own machinery.

The bridge-valence diagnostic is the answer to one question:

$$\boxed{\text{What must be read out when a relation-first programme audits a bridge claim?}}$$

A bridge claim has the form:

$$X - \mathfrak{B} - Y.$$

By No Undeclared Relation, the bridge is inadmissible unless the relevant boundary, mediation, and observer/readout faces are declared. By Recursive Bridge Calibration, the bridge also needs a target distinction family, evaluator, admission rule, tolerance, and update rule. Therefore the diagnostic must record exactly the quantities that determine whether a bridge is bounded, finitely unifying, critical, underdeclared, or mixed.

1A.1 Derivation of ρ_{face}: face-demand rank

A target distinction is never free-floating. It requires relation faces:

$$(\mathsf{Z}, \mathsf{R}, \mathsf{O}).$$

Thus, when a target D'_Y is declared, the first invariant is:

$$\boxed{\rho_{\text{face}} = \text{number of independent target-face demands not already recovered by declared bridge laws.}}$$

This is forced by:

- Relation-to-Triad Declaration;
- No Undeclared Boundary;
- No Undeclared Observer;
- Closure-incomplete singleton/dyad no-gos.

If a target activates one coherent face family, the bridge tends to be bounded. If it activates many independent face families, bridge-valence rises.

1A.2 Derivation of b_{targ}: target bridge-valence lower bound

Bridge accessibility says that a target is accessible only through an admitted bridge recovering declared target distinctions:

$$D'_Y \subseteq \mathrm{Rec}_{\mathfrak{B}}(D_X).$$

If the target family decomposes into several non-equivalent target presentations, each requires either a bridge or a declared reduction to another bridge.

Thus:

$$\boxed{b_{\text{targ}} = \text{minimum number of target-presentation branches demanded by } D'_Y.}$$

This is forced by:

- bridge accessibility;
- No-Free-Transfer;
- target-presentation calibration;
- closure-atlas branch selection.

1A.3 Derivation of s_{sub}: substitution depth

Triadic substitution says that a nontrivial role substitute must itself be triadically declared.

If a target uses a role like “observer,” “volume,” “region,” “dimension,” “dynamics,” or “empirical readout,” and that role performs nontrivial work, then the role opens into its own boundary–mediation–observer structure.

Thus:

$$\boxed{s_{\text{sub}} = \text{maximum recursive depth of nontrivial role openings required by the target.}}$$

This is forced by:

- Triadic Substitution Principle;
- role-recursion / substitution recursion;
- no dangling role interface;
- Recursive Chart Learning update structure.

1A.4 Derivation of n_{NFT}: no-free-transfer count

Every time a bridge tries to move from one structure type to another without a declared transfer law, No-Free-Transfer activates.

Examples:

$$\text{order} \not\Rightarrow \text{dimension};$$

$$\text{counting} \not\Rightarrow \text{continuum volume};$$

$$\text{metric} \not\Rightarrow \text{dynamics};$$

$$\text{recoverability} \not\Rightarrow \text{causality}.$$

Thus:

$$\boxed{n_{\mathrm{NFT}} = \text{number of target-type jumps not already supplied by declared bridge laws.}}$$

This is forced by:

- No-Free-Transfer;
- structure-type no-promotion;
- task-visible quotient admissibility;
- bridge-accessibility semantics.

1A.5 Derivation of κ_{compat}: compatibility-operator availability

The programme already has valued face-recovery machinery:

$$\mathsf{O} = \mathsf{Z} \otimes \mathsf{R},$$

and residual/quotient operations:

$$\mathsf{Z} = \mathsf{O} \oslash_Z \mathsf{R}, \quad \mathsf{R} = \mathsf{Z} \oslash_R \mathsf{O}.$$

When a compatibility operator, field equation, symmetry, transform, or bridge law collapses multiple target demands into one finite system, a high-valence problem can become a finite unification atlas rather than a critical locus.

Thus:

$$\boxed{\kappa_{\mathrm{compat}} = \text{availability and strength of declared compatibility operators collapsing target demands.}}$$

This is forced by:

- valued face-recovery;
- residual algebra;
- compatibility law;
- finite unification via common operator.

1A.6 Derivation of h_{obs}: observer-slice admissibility hardness

No Undeclared Observer says no visibility-bearing inference is legitimate without a declared observer/readout/evaluator.

Some targets have simple readout. Others require empirical measurement, chart-stable region presentation, probability update, first-person/third-person alignment, or physical instrumentation.

Thus:

$$\boxed{h_{\mathrm{obs}} = \text{observer/readout/evaluator/calibrator burden required by the target.}}$$

This is forced by:

- No Undeclared Observer;
- Charted Organization;
- Recursive Chart Learning;
- Recursive Bridge Calibration;
- empirical-readout residuals.

Theorem 1A.7 — Programme-Native Diagnostic Derivation Theorem

The six pre-diagnostic coordinates:

$$(\rho_{\text{face}}, b_{\text{targ}}, s_{\text{sub}}, n_{\text{NFT}}, \kappa_{\text{compat}}, h_{\text{obs}})$$

are forced by relation-first bridge audit. They are not independent external scoring choices.

Proof

A bridge claim must declare target faces, target distinctions, bridge relation, role substitutions, target-type transfers, compatibility/recovery laws, observer/readout, and calibrator. Each required item corresponds to one diagnostic coordinate.

The target faces produce ρ_{face}. The number of target-presentation branches produces b_{targ}. Nontrivial role openings produce s_{sub}. Type jumps without declared bridge laws produce n_{NFT}. Declared compatibility/recovery operators produce κ_{compat}. Observer/readout/evaluator/calibrator burden produces h_{obs}.

Since these are exactly the quantities a relation-first audit must inspect, the diagnostic vector is programme-native. □

Corollary 1A.8 — The rubric is a readout, not an imposition

The diagnostic rubric is a chart/readout of the bridge-audit state. It does not impose a foreign taxonomy on the programme.

Corollary 1A.9 — Countability and chartability of the diagnostic

For a finite declared target family D'_Y, finite target columns, and finite bridge-audit vocabulary, the diagnostic is countable and chartable:

$$\mathcal{D}_{\text{pre}}(X, D'_Y) = (\rho_{\text{face}}, b_{\text{targ}}, s_{\text{sub}}, n_{\text{NFT}}, \kappa_{\text{compat}}, h_{\text{obs}})$$

is a finite readout tuple.

When the target family is infinite or open-ended, the diagnostic must be localized to a declared finite or recursively generated target subfamily. Otherwise it returns insufficiently declared or diagnostic-pending status.

This follows from No Undeclared Boundary: a diagnostic cannot classify an undeclared totality.

1B. Programme-native recursion of the diagnostic

The previous section derives the six diagnostic coordinates. But the diagnostic is still incomplete if treated as a single flat readout.

In the programme, residuals do not merely sit at the end of a failed proof. Residuals generate update, learning, and branch-field change. Therefore the diagnostic must be recursive.

This is not an extra design choice. It follows from existing programme commitments in the paper stack:

1. **Recursive Chart Learning:** an admitted residual/update changes later available branches;
2. **Recursive Bridge Calibration:** a failed or partial bridge records an obstruction and updates the next bridge choice;
3. **Triadic recursion:** residual, trace, and substitution recursion are the three face-dominant modes of recursive relation;
4. **Closure Atlas / coupling machinery:** residuals activate landed or stranded branch interfaces;

5. **No Undeclared Relation / Boundary / Observer:** a residual cannot be treated as resolved until its required faces are declared.

The internal audit internal workflow note can check whether these commitments have been applied, but it is not part of the paper's theorem-source.

1B.1 Programme theorem/no-go tree

Let:

$$\mathcal{T}_{\text{prog}}$$

be the programme theorem/no-go tree. It is not a primitive object. It is the organised relation-presentation of available theorem, no-go, residual, chart, trace, quotient, and calibrator machinery.

At minimum it contains subtrees:

Residual / issue	Activated subtree
Δ_{decl}	NUR / NUB / NUO
Δ_{face}	Triadic Closure / six unstable subconfiguration no-gos
Δ_{sub}	Triadic Substitution / role-recursion
Δ_{trace}	Recursive Chart Learning / no dangling path trace
Δ_{chart}	Charted Organization / gluing / holonomy
Δ_{quot}	task-visible quotient admissibility
Δ_{type}	structure-type no-promotion
Δ_{cal}	RBC / calibrator boundedness
Δ_{compat}	valued face-recovery / compatibility operator
Δ_{emp}	observer/readout / empirical calibration

Definition 1B.2 — Diagnostic state

A diagnostic state at pass n is:

$$\boxed{\mathfrak{D}_n = (\mathcal{D}^{(n)}_{\text{pre}}, \mathcal{D}^{(n)}_{\text{post}}, \Delta^{(n)}_{\text{diag}}, \mathcal{A}_n, \mathcal{B}_n, \text{Status}_n).}$$

Where:

- $\mathcal{D}^{(n)}_{\text{pre}}$ is the current pre-diagnostic vector;
- $\mathcal{D}^{(n)}_{\text{post}}$ is the current post-diagnostic vector;
- $\Delta^{(n)}_{\text{diag}}$ is the current diagnostic residual;
- $\mathcal{A}_n \subseteq \mathcal{T}_{\text{prog}}$ is the active theorem/no-go subtree;
- $\mathcal{B}_n$ is the current branch field;
- Status_n is the current diagnostic status.

Definition 1B.3 — Activation map

The activation map:

$$\text{Act} : \Delta^{(n)}_{\text{diag}} \to \mathcal{P}(\mathcal{T}_{\text{prog}})$$

selects the theorem/no-go subtree relevant to the current residual.

For example:

$$\mathrm{Act}(\Delta_{\mathrm{quot}}) = \{\text{task-visible quotient admissibility}\},$$

and:

$$\mathrm{Act}(\Delta_{\mathrm{face}}) = \{\text{Triadic Closure, six subconfiguration no-gos}\}.$$

Definition 1B.4 — Recursive diagnostic update

The recursive diagnostic update is:

$$\boxed{\mathfrak{D}_{n+1} = U_{\mathrm{diag}}(\mathfrak{D}_n, \Omega_n, \mathcal{T}_{\mathrm{prog}}).}$$

where Ω_n is the residual family visible at pass n.

The update performs four operations:

1. activates the relevant theorem/no-go subtree;
2. applies the subtree to refine the diagnostic residual;
3. updates the branch field $\mathcal{B}_n$;
4. returns a new diagnostic status.

Definition 1B.5 — Extended diagnostic residual algebra

The diagnostic residual lives in an extended residual algebra:

$$\boxed{\Delta_{\mathrm{diag}} \in \mathcal{E}^{\top}_{\mathrm{diag}}.}$$

The top element:

$$\top$$

means inadmissible, undefined, or blocked by a no-go.

The important outcomes are:

Outcome	Residual behavior	Meaning
closure	$\Delta_{\mathrm{diag}} \to 0$	classification closes
blockage	$\Delta_{\mathrm{diag}} = \top$	a no-go blocks the branch
bounded residualization	$0 < \Delta_{\mathrm{diag}} \preceq \varepsilon_{\mathrm{diag}}$	conditional / approximate / residualized classification
persistent branching	$\Delta_{\mathrm{diag}} \not\to 0$, $\Delta_{\mathrm{diag}} \neq \top$, stable residual family recurs	critical-locus signature

Theorem 1B.6 — Recursive Diagnostic Necessity Theorem

If a relation-first diagnostic produces a nonzero residual, then the diagnostic must either:

1. close the residual by applying already declared theorem machinery;
2. block the branch by an active no-go;
3. bound/residualize the branch under a declared calibrator; or
4. update the branch field through a recursive diagnostic pass.

It may not leave the residual inert.

Proof

A diagnostic residual is a declared failure, gap, or underdetermination in a bridge-classification relation. By No Undeclared Boundary, the failure boundary must be declared. By No Undeclared Observer, it must be readable/evaluable. By Recursive Bridge Calibration, a failed or partial bridge generates an update obligation. By Recursive Chart Learning, an admitted residual/update changes later available branch structure. Therefore a nonzero diagnostic residual must either close, block, bound, or update the diagnostic state. Leaving it inert would violate the programme's residual-to-update discipline. □

Theorem 1B.7 — Theorem/No-Go Tree Activation Theorem

For every named diagnostic residual Δ_i, there is a programme-native theorem/no-go subtree that must be activated before the diagnostic status can be admitted.

Proof

A named diagnostic residual identifies a missing or failed component of the bridge relation: declaration, face structure, substitution, trace, chart, quotient, type, calibration, compatibility, or empirical readout. Each of these components is governed by an existing programme theorem/no-go family. If the relevant subtree is not activated, then the diagnostic has not inspected the residual under the machinery that defines it. The diagnostic status is therefore underdeclared. Thus each named residual forces activation of its theorem/no-go subtree. □

Corollary 1B.8 — The diagnostic is learning-theoretic

The bridge-valence diagnostic is an instance of programme-native learning:

$$\text{question} \to \text{answer attempt} \to \text{residual} \to \text{theorem/no-go activation} \to \text{branch-field update.}$$

Corollary 1B.9 — Persistent branching is not failure to halt

If the recursive diagnostic does not converge to zero and does not hit $\top$, but stabilizes into a finite recurring residual family that generates stable bridge obligations, then the appropriate output is not "failed diagnostic." It is:

$$\boxed{\text{critical relation locus / branching target atlas.}}$$

Register note

This recursive layer should not be used to dump every theorem into every diagnostic case. The activation map selects only the subtree relevant to the current residual. That is what keeps the diagnostic repeatable.

2. Target-demand matrix

Let X be a source relation-presentation and D'_Y a declared target distinction family.

Definition 2.1 — Target-demand matrix

The **target-demand matrix** of a bridge problem is:

$$\mathsf{T}(X, D'_Y) = \begin{array}{c|c} & \begin{array}{cccccccc} \prec & \mathcal{I} & \tau & \mathcal{C} & d & \mu/\text{scale} & \text{dyn} & \text{emp} \end{array} \\ \hline \begin{array}{c} \mathsf{Z} \\ \mathsf{R} \\ \mathsf{O} \end{array} & \begin{bmatrix} z_1 & z_2 & z_3 & z_4 & z_5 & z_6 & z_7 & z_8 \\ r_1 & r_2 & r_3 & r_4 & r_5 & r_6 & r_7 & r_8 \\ o_1 & o_2 & o_3 & o_4 & o_5 & o_6 & o_7 & o_8 \end{bmatrix} \end{array}.$$

The columns are target distinction types. The rows are relation-first faces:

$$(\mathsf{Z}, \mathsf{R}, \mathsf{O}).$$

An entry is active when the target distinction requires that face to be declared and preserved.

The listed columns are not exhaustive; they are the common columns for the current AC/physics-adjacent branch.

Definition 2.2 — Face-demand rank

The **face-demand rank**:

$$\rho_{\text{face}}(X, D'_Y)$$

is the number of independent face-columns in T that cannot be recovered from one another under already-declared bridge laws.

This is not ordinary linear rank unless a vector-space semantics is declared. It is a relation-first independence count.

Definition 2.3 — Target bridge-valence lower bound

The **target bridge-valence lower bound**:

$$b_{\text{targ}}(X, D'_Y)$$

is the number of distinct target-presentation classes activated by the target family.

Examples of target-presentation classes include:

1. order/preorder;
2. interval-region;
3. topology/locale;
4. dimension;
5. scale/measure;
6. dynamics/update;
7. theorem-specific reconstruction;
8. empirical readout.

Definition 2.4 — Substitution depth

The **substitution depth**:

$$s_{\text{sub}}$$

is the maximum number of times a role used as a face must itself be opened as a nontrivial triadic subproblem.

For example, "observer-volume readout" may open into:

$$\text{observer} \to (\text{chart}, \text{region task}, \text{volume readout}, \text{calibrator}).$$

Definition 2.5 — No-free-transfer count

The **no-free-transfer count**:

$$n_{\mathrm{NFT}}$$

is the number of target-presentation transfers required but not already supplied by source structure.

For example:

$$\text{order} \not\Rightarrow \text{dimension},$$

$$\text{counting} \not\Rightarrow \text{continuum volume},$$

$$\text{metric} \not\Rightarrow \text{dynamics}.$$

Each such target jump increments n_{NFT} unless a bridge law is already declared.

Definition 2.6 — Compatibility-operator availability

$$\kappa_{\mathrm{compat}}$$

records whether the target family has a known compatibility operator, law, or structure that collapses multiple target demands into one bounded bridge.

Examples include:

- a dynamical equation relating force and acceleration;
- field equations relating electric/magnetic fields and sources;
- gauge symmetry plus representation structure constraining interactions;
- a declared bridge law recovering target distinctions.

Definition 2.7 — Observer-slice admissibility hardness

$$h_{\mathrm{obs}}$$

measures how much of the target family must survive the same observer-slice boundary/readout/calibrator.

A problem is more critical when many target directions remain admissible or conditionally admissible under one slice.

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3. Pre-solution diagnostic

Definition 3.1 — Pre-solution diagnostic vector

Define:

$$\boxed{\mathcal{D}_{\mathrm{pre}}(X, D'_Y) = (\rho_{\mathrm{face}}, b_{\mathrm{targ}}, s_{\mathrm{sub}}, n_{\mathrm{NFT}}, \kappa_{\mathrm{compat}}, h_{\mathrm{obs}}).}$$

Definition 3.2 — Bounded bridge prediction

A problem is predicted to be a **bounded bridge** when:

1. ρ_{face} is low;
2. b_{targ} is low;
3. $s_{\text{sub}} \leq 1$;
4. n_{NFT} is low;
5. κ_{compat} is present;
6. h_{obs} is low or moderate.

The expected closure form is:

$$\boxed{X - \mathfrak{B} - Y}$$

with a finite residual vector.

Definition 3.3 — Finite unification atlas prediction

A problem is predicted to be a **finite unification atlas** when:

1. ρ_{face} and b_{targ} are high;
2. a strong compatibility operator κ_{compat} is present;
3. the operator collapses multiple bridge families into one finite equation-system, symmetry, or compatibility law;
4. residuals outside the declared target remain but do not prevent closure inside the target.

The expected closure form is:

$$\boxed{\{\mathfrak{B}_i : X_i \to Y_i\}_{i=1}^{k} \quad \text{collapsed by} \quad \kappa_{\text{compat}}.}$$

Definition 3.4 — Critical relation locus prediction

A problem is predicted to be a **critical relation locus** when:

1. ρ_{face} is high;
2. b_{targ} is high;
3. $s_{\text{sub}} \geq 2$;
4. n_{NFT} is high;
5. no single compatibility operator collapses all target demands;
6. many bridge directions remain admissible or conditionally admissible under one observer slice;
7. target completion repeatedly opens new target-presentation branches.

The expected closure form is not terminal:

$$\boxed{\Delta = 0}$$

but atlasic:

$$\boxed{\mathfrak{A}_X^{\text{branch}} = \{\mathfrak{B}_i : X \to Y_i\}_{i \in I}.}$$

Definition 3.5 — Accidental underdeclaration prediction

A problem is predicted to be **accidental underdeclaration** when apparent high residuality is caused by missing source/target typing, missing boundary, missing observer/readout, omitted theorem, or object-language smuggling.

Such residuals should disappear after diagnostic audit.

4. Post-diagnostic

Definition 4.1 — Residual persistence score

The **residual persistence score**:

$$r_{\text{persist}}$$

records whether the same residual types recur after:

1. relation-first language audit;
2. source–bridge–target audit;
3. diagnostic audit;
4. comparison across multiple target presentations.

Definition 4.2 — Bridge-generation score

The **bridge-generation score**:

$$g_{\text{bridge}}$$

records whether persistent residuals generate stable next-bridge families.

Definition 4.3 — Diagnostic-survival score

The **diagnostic-survival score**:

$$a_{\text{diag}}$$

records whether residuals survive no-go, quotient, chart, trace, calibrator, and target-type checks.

Definition 4.4 — Post-diagnostic confirmation

The post-diagnostic is:

$$\boxed{\mathcal{D}_{\text{post}} = (r_{\text{persist}}, g_{\text{bridge}}, a_{\text{diag}}).}$$

A predicted critical locus is confirmed only when all three are high.

4A. Diagnostic error, abstention, and borderline cases

The bridge-valence diagnostic is not a total classifier.

A diagnostic that always classifies would violate the programme's own boundary discipline: if the source, target, observer slice, or comparison task is underdeclared, then the diagnostic itself is underdeclared.

Definition 4A.1 — Diagnostic-error vector

The **diagnostic-error vector** is:

$$\boxed{\Delta_{\mathrm{diag}} = \Delta_{\mathrm{decl}} \oplus \Delta_{\mathrm{face}} \oplus \Delta_{\mathrm{val}} \oplus \Delta_{\mathrm{sub}} \oplus \Delta_{\mathrm{NFT}} \oplus \Delta_{\kappa} \oplus \Delta_{\mathrm{obs}} \oplus \Delta_{\mathrm{hist}}.}$$

where:

Component	Meaning
Δ_{decl}	source, target, task, or boundary is underdeclared
Δ_{face}	face-demand rank cannot be determined
Δ_{val}	bridge-valence lower bound is uncertain
Δ_{sub}	substitution depth is uncertain
Δ_{NFT}	no-free-transfer count is uncertain
Δ_{κ}	compatibility-operator availability is unknown or ambiguous
Δ_{obs}	observer-slice admissibility is uncertain
Δ_{hist}	historical/scientific calibration case is too coarse or contested

Definition 4A.2 — Diagnostic status values

The diagnostic may return one of the following statuses:

Status	Meaning
admitted classification	sufficient declaration and diagnostic margin
conditional classification	classification holds under declared additional hypotheses
borderline classification	two or more classes have small margin separation
mixed classification	different subtargets belong to different classes
insufficiently declared	source/target/faces/calibrator not declared enough to classify
classification blocked	a no-go prevents the proposed class assignment
diagnostic pending	post-diagnostic confirmation has not yet been run

Table-status convention

The diagnostic landscape table is schematic, so it sometimes appends qualifiers to the base statuses. For example:

$$extadmittedschematic$$

means:

$$extadmittedclassificationforthedeclaredschematictargetregime.$$

Likewise:

$$extconditionalschematic$$

means:

$$extconditionalclassificationforthedeclaredschematictargetregime.$$

Phrases such as “mixed / conditional,” “borderline / programme-dependent,” or “target-dependent” are not extra theorem classes. They are table-level qualifiers of the base statuses: mixed, borderline, conditional, or underdetermined.

Definition 4A.3 — Diagnostic margin

Let $S_C(X, D)$ be the diagnostic score for class C. The **classification margin** for candidate class C is:

$$\boxed{m_C(X, D) = S_C(X, D) - \max_{C' \neq C} S_{C'}(X, D).}$$

No specific numerical scoring rule is assumed by default. A project may use ordinal, Boolean, weighted, or quantitative scoring, provided the scoring rule is declared.

Definition 4A.4 — Admitted diagnostic classification

A classification:

$$\text{Class}(X, D) = C$$

is **admitted** only if:

1. the source and target are sufficiently declared:
$$\Delta_{\text{decl}} \preceq \varepsilon_{\text{decl}};$$
2. the diagnostic error is bounded:
$$\Delta_{\text{diag}} \preceq \varepsilon_{\text{diag}};$$
3. the class margin is positive beyond tolerance:
$$m_C(X, D) > \eta_{\text{class}};$$
4. post-diagnostic confirmation agrees with the pre-diagnostic or supplies a declared correction.

If these conditions fail, the diagnostic must abstain, return a conditional classification, or return a mixed/borderline classification.

Theorem 4A.5 — Bridge-Valence Classification is Partial and Calibrated

The bridge-valence diagnostic is a partial calibrated classifier. It does not assign an admitted class to every bridge problem.

Proof

An admitted classification is itself a relation-first claim: it relates a source-target bridge problem to a diagnostic class. Therefore it requires declared boundary, mediation, observer/readout, and calibrator conditions. If the source, target, diagnostic dimensions, scoring rule, observer slice, or post-diagnostic status is underdeclared, then the classification relation is underdeclared. By No Undeclared Relation, no admitted classification follows. If two classes are within margin tolerance, the diagnostic cannot distinguish them to the declared precision. Therefore the diagnostic is partial and calibrated rather than total. □

Corollary 4A.6 — No forced classification

If:

$$\Delta_{\text{diag}} \not\preceq \varepsilon_{\text{diag}}$$

or:

$$m_C(X, D) \leq \eta_{\text{class}},$$

then the diagnostic must not force X into class C. The correct output is underdetermined, borderline, mixed, conditional, or diagnostic-pending.

Corollary 4A.7 — Compatibility-operator discoveries can change classification

If a new compatibility operator:

$$\kappa_{\text{compat}}$$

is discovered or declared, a problem previously diagnosed as critical may become a finite unification atlas.

This is not diagnostic failure. It is an update of Δ_κ and the target-demand matrix.

Maxwellian electromagnetism is the schematic example: pre-unification electric/magnetic phenomena may appear crossroads-structured; the field-equation compatibility law stabilizes them into a finite unification atlas.

5. Diagnostic theorem

Theorem 5.1 — Error-Bounded Bridge-Valence Diagnostic Theorem

Let $X - D'_Y$ be a target-bridge problem in an observer slice.

The diagnostic returns:

$$\text{Diag}(X, D'_Y) = (C, \Delta_{\text{diag}}, m_C, \text{Status}).$$

If:

1. $\Delta_{\text{diag}} \preceq \varepsilon_{\text{diag}}$;
2. $m_C(X, D'_Y) > \eta_{\text{class}}$;
3. pre-diagnostic and post-diagnostic agree, or their disagreement is resolved by declared update;

then the diagnostic may assign admitted class C.

The admitted class is determined as follows:

- If $\mathcal{D}_{\text{pre}}$ predicts bounded bridge and $\mathcal{D}_{\text{post}}$ shows low residual persistence after diagnostic audit, then $C =$ bounded bridge.
- If $\mathcal{D}_{\text{pre}}$ predicts finite unification atlas and a compatibility operator κ_{compat} collapses the target-demand branches into a finite admitted law/system, then $C =$ finite unification atlas.
- If $\mathcal{D}_{\text{pre}}$ predicts critical locus and $\mathcal{D}_{\text{post}}$ confirms residual persistence, bridge generation, and audit survival, then $C =$ critical relation locus.

If the error bound or margin condition fails, the diagnostic must return a non-admitted status: under-declared, borderline, mixed, conditional, blocked, or diagnostic-pending.

Proof

The definitions of bounded bridge, finite unification atlas, and critical relation locus are based on face-demand complexity, bridge-valence, substitution depth, no-free-transfer count, compatibility-law availability, and observer-slice admissibility. But classification itself is a relation-first claim, so it must be admitted by a diagnostic calibrator. The error bound ensures the diagnostic inputs are sufficiently declared. The margin condition ensures the selected class is distinguishable from alternatives. Post-diagnostic agreement ensures the classification survives residual and no-go checks. When these conditions hold, the class assignment is admitted. When they fail, No Undeclared Relation and calibration discipline block forced classification. □

5A. Actionability of a critical-locus classification

A critical-locus classification is not a label for difficulty and not a declaration of permanent unsolvability.

It has constructive content.

It changes:

1. the admissible answer-space;
2. the appropriate research method;
3. the success criterion;
4. the falsification/downgrade target.

Definition 5A.1 — Critical-locus action protocol

If a target-bridge problem $X - D'_Y$ is classified as a critical relation locus, the appropriate response is the two-track protocol:

$$\boxed{\mathrm{Act}_{\mathrm{crit}} = (\mathrm{AtlasMap}, \mathrm{CompatSearch}).}$$

Track 1 — Atlas articulation

$$\mathrm{AtlasMap}(X, D'_Y)$$

requires:

1. identify the distinct target presentations admitted by D'_Y;
2. decompose each target presentation into sub-bridges;
3. classify each sub-bridge as bounded, finite, conditional, residualized, blocked, or critical;
4. record the residual profile of each target presentation;
5. identify admissible partial closures;
6. record target-presentation dependencies and incompatibilities.

Track 2 — Compatibility-operator search

$$\mathrm{CompatSearch}(X, D'_Y)$$

requires:

1. identify which diagnostic coordinates cause criticality;
2. ask what operator, theorem, target re-declaration, or bridge law would reduce them;
3. search for a compatibility operator:
   $$\kappa^*_{\mathrm{compat}}$$
   capable of collapsing the active target faces to declared tolerance;
4. if such an operator is found, rerun the diagnostic;
5. if no such operator is declared, continue atlas articulation without pretending terminal closure has been achieved.

Definition 5A.2 — Coordinate repair targets

For a critical locus with:

$$(\rho, b, s, n, \kappa, h),$$

the coordinate repair targets are:

High coordinate	Repair target
ρ_{face} high	collapse independent target-face demands by a shared compatibility law
b_{targ} high	identify target presentations that reduce to a common branch
s_{sub} high	supply role-substitution closures that stop recursive opening
n_{NFT} high	supply declared transfer laws for type jumps
$\kappa_{\text{compat}} = N/P$	discover or declare a terminal or partial compatibility operator
h_{obs} high	declare observer/readout/evaluator/calibrator structures that reduce readout burden

Theorem 5A.3 — Critical-Locus Method-Matching Theorem

If a problem is diagnosed as a critical relation locus, then bounded-bridge methodology is inadmissible as a complete method for that problem at the declared target regime. The appropriate complete method is the two-track critical-locus protocol:

$$\boxed{\text{AtlasMap} + \text{CompatSearch}\,.}$$

Proof

A bounded bridge requires low target-valence, low no-free-transfer burden, low substitution depth, and a declared compatibility/admission path sufficient for a single target closure. A critical relation locus, by definition, has high face-demand, high target-valence, deep substitution, high no-free-transfer count, no terminal compatibility operator, and persistent residuals under diagnostic audit. Therefore a single bounded bridge does not recover the declared target family. Atlas articulation is required to record the target branches and residual profiles. Compatibility-operator search is required because the classification can be downgraded if a terminal compatibility operator is discovered. Thus the complete method is the two-track protocol. □

Corollary 5A.4 — Progress criterion for critical loci

For a bounded bridge, progress is convergence to a single target closure.

For a finite unification atlas, progress is discovery or refinement of the compatibility operator.

For a critical relation locus, progress is:

$$\boxed{\text{atlas completeness} + \text{admitted partial closures} + \text{localized residuals} + \text{compatibility-operator search.}}$$

Corollary 5A.5 — Criticality is not permanent

If a compatibility operator:

$$\kappa^*_{\text{compat}}$$

is discovered that collapses all active target faces to declared tolerance, then the classification downgrades:

$$\boxed{\text{critical locus} \rightsquigarrow \text{finite unification atlas.}}$$

This is a diagnostic update, not a contradiction.

Corollary 5A.6 — Partial closures are genuine outputs

A critical locus may contain many bounded or finite sub-bridges. These are real research outputs. They do not close the entire critical locus, but they populate its branching atlas.

Register note

The two-track protocol prevents two opposite mistakes:

1. treating the atlas itself as a reason to stop searching for compatibility operators;
2. treating compatibility-operator search as a reason to ignore the atlas structure already visible.

5B. Critical-locus action space

The previous section states the critical-locus action protocol. This section derives the action space in which that protocol lives.

The key point is:

$$\boxed{\mathrm{Act}_{\mathrm{crit}} \text{ is not an extra heuristic. It is a generated subspace of the diagnostic action space.}}$$

Definition 5B.1 — Diagnostic action

Let:

$$\mathfrak{D}_n = (\mathcal{D}^{(n)}_{\mathrm{pre}}, \mathcal{D}^{(n)}_{\mathrm{post}}, \Delta^{(n)}_{\mathrm{diag}}, \mathcal{A}_n, \mathcal{B}_n, \mathrm{Status}_n)$$

be a diagnostic state.

A **diagnostic action** is an admitted update:

$$a : \mathfrak{D}_n \to \mathfrak{D}_{n+1}$$

that is generated by one of the programme-native operations:

1. triadic declaration;
2. charted readout refinement;
3. target-presentation split;
4. bridge-localization;
5. residual profiling;
6. compatibility-operator search or declaration;
7. calibrator declaration/refinement;
8. theorem/no-go activation;
9. trace/update admission.

The diagnostic action space is:

$$\boxed{\mathcal{U}_{\mathrm{diag}}(\mathfrak{D}_n) = \{a : \mathfrak{D}_n \to \mathfrak{D}_{n+1} \text{ admitted by the diagnostic calibrator}\}.}$$

Definition 5B.2 — Critical diagnostic state

A diagnostic state is **critical** when:

$$\text{Status}_n = \text{critical}$$

and the pre/post diagnostic data include:

1. high face-demand:
$$\rho_{\text{face}} = H \text{ or } VH;$$
2. high target-valence:
$$b_{\text{targ}} = H \text{ or } VH;$$
3. nontrivial substitution depth:
$$s_{\text{sub}} \geq M;$$
4. high no-free-transfer burden:
$$n_{\text{NFT}} = H \text{ or } VH;$$
5. absent or partial terminal compatibility:
$$\kappa_{\text{compat}} = N \text{ or } P;$$
6. persistent residuals:
$$r_{\text{persist}} > 0;$$
7. bridge-generation pressure:
$$g_{\text{bridge}} > 0.$$

Definition 5B.3 — Critical action subspace

For a critical diagnostic state $\mathfrak{D}$, the **critical action subspace** is:

$$\boxed{\mathcal{U}_{\text{crit}}(\mathfrak{D}) \subseteq \mathcal{U}_{\text{diag}}(\mathfrak{D})}$$

consisting of diagnostic actions that either:

1. articulate the branching target atlas; or
2. search for / declare compatibility operators that could reduce the criticality.

Thus:

$$\boxed{\mathcal{U}_{\text{crit}} = \mathcal{U}_{\text{atlas}} \cup \mathcal{U}_{\kappa}.}$$

where:

$$\mathcal{U}_{\text{atlas}} = \{\text{target split, bridge localization, residual profiling, partial closure admission}\},$$

and:

$$\mathcal{U}_{\kappa} = \{\text{compatibility-operator search, operator declaration, target-collapse test}\}.$$

Definition 5B.4 — Atlas-action generators

The atlas-action generators are:

Generator	Action
SplitTarget	split D'_Y into target presentations
LocalizeBridge	assign each target presentation a bridge candidate
ProfileResidual	record residual vector for each branch
AdmitPartial	admit bounded/finite subbridges where they close
MapDependencies	record downstream dependencies and incompatibilities
UpdateAtlas	update the branch atlas after each closure/blockage

Collectively these generate:

$$\mathrm{AtlasMap}\,.$$

Definition 5B.5 — Compatibility-search generators

The compatibility-search generators are:

Generator	Action
FindKappa	search for a candidate compatibility operator
DeclareKappa	declare its domain, range, tolerance, and target faces
TestCollapse	test whether target branches collapse under κ
DowngradeClass	update critical to finite unification if collapse succeeds
RejectKappa	record failed operator and residual
RefineTarget	modify target regime if the operator only works locally

Collectively these generate:

$$\mathrm{CompatSearch}\,.$$

Theorem 5B.6 — Critical Action-Space Generation Theorem

For any critical diagnostic state $\mathfrak{D}$, the minimal non-inert admissible action space contains:

$$\boxed{\mathrm{AtlasMap} + \mathrm{CompatSearch}\,.}$$

Proof

A critical diagnostic state has high target-valence and persistent residuals. High target-valence means the target cannot be treated as a single bounded branch without overclaim. Therefore target-presentation splitting, bridge localization, and residual profiling are forced. These are exactly the atlas-action generators.

The same critical state has absent or partial terminal compatibility. Since a terminal compatibility operator would downgrade the classification to finite unification, the diagnostic must search for or test candidate compatibility operators before treating criticality as stable. These are exactly the compatibility-search generators.

If neither atlas-action nor compatibility-search actions are taken, the residual is left inert. That violates Recursive Diagnostic Necessity. Therefore the minimal non-inert critical action space contains $\mathrm{AtlasMap} + \mathrm{CompatSearch}$. □

Corollary 5B.7 — Protocol/action-space relation

The two-track protocol is not the whole action space. It is the minimal generating form of the critical action subspace:

$$\boxed{\mathrm{Act}_{\mathrm{crit}} = (\mathrm{AtlasMap}, \mathrm{CompatSearch}) \subseteq \mathcal{U}_{\mathrm{crit}}.}$$

The protocol tells the researcher which kinds of actions are admissible first. The action space lists the available operations inside those kinds.

Corollary 5B.8 — No single-answer action without downgrade

A single-answer closure action is admissible for a critical locus only after a compatibility-search action succeeds and downgrades the classification.

Until then, single-answer closure is an overclaim.

Corollary 5B.9 — Action-space revision

If the programme discovers new theorem machinery, the action space may expand. Such expansion must occur through:

$$\mathrm{Update}_{\mathrm{diag}}$$

and must be recorded as a diagnostic revision.

Register note

This section answers a possible objection: "why these actions?"

Answer:

$$\boxed{\text{because criticality is exactly the diagnostic state in which target-branching and compatibility failure are both a}}$$

So atlas mapping and compatibility search are not choices by taste. They are the two forced non-inert responses to the two active structural facts.

6. Schematic calibration cases

The following cases are schematic calibration examples, not detailed historical reconstructions. Their role is to show that the diagnostic can classify familiar bridge problems without assigning "criticality" merely because a problem is difficult.

Each case should be read with diagnostic-error caution: if the target regime is changed, the classification may change.

6.1 Newtonian mechanics as bounded bridge inside declared regime

Target

Given masses, positions, forces, and inertial-frame structure, recover accelerations/trajectories.

Diagnostic

- ρ_{face}: moderate;
- b_{targ}: low;
- s_{sub}: low;
- n_{NFT}: low inside the declared regime;
- κ_{compat}: present through equations of motion;
- h_{obs}: moderate.

Classification

$$\boxed{\text{bounded bridge inside declared Newtonian regime.}}$$

Caveat

This does not mean Newtonian mechanics is globally terminal. Relativity and quantum theory activate targets outside the declared Newtonian boundary.

6.2 Maxwellian electromagnetism as finite unification atlas

Target

Unify electric, magnetic, current/source, and wave/optical phenomena under field equations.

Diagnostic

- ρ_{face}: high;
- b_{targ}: high;
- s_{sub}: moderate;
- n_{NFT}: initially high;
- κ_{compat}: present through Maxwell field equations and field/potential structure;
- h_{obs}: high but stabilized.

Classification

$$\boxed{\text{crossroads-structured but finitely unified.}}$$

Maxwellian theory is not low-valence. It is a finite unification atlas because the field equations provide a common compatibility law collapsing several bridge families into one system.

6.3 Standard Model gauge structure as high-valence finite unification atlas

Target

Organize strong, weak, electromagnetic interactions and matter representations under gauge symmetry, field content, and interaction terms.

Diagnostic

- ρ_{face}: high;
- b_{targ}: high;
- s_{sub}: high;
- n_{NFT}: high before gauge/representation structure;
- κ_{compat}: present through gauge symmetry, representation assignments, renormalizable interaction structure, and Higgs mechanism;
- h_{obs}: high.

Classification

$$\boxed{\text{high-valence finite unification atlas within declared quantum-field-theoretic target.}}$$

Caveat

Residuals such as gravity, dark matter, hierarchy, neutrino mass details, or quantum-gravity issues are not failures of the bounded SM target. They are outside or at the boundary of that declared target.

6.4 Observer-slice AC / continuum node as critical relation locus

Target

Connect observer-slice AC presentation to AC-readiness, causal-set-style, Lorentzian, theorem-specific, metric-scale, and GR-facing targets.

Diagnostic

- ρ_{face}: high;
- b_{targ}: high;
- s_{sub}: high;
- n_{NFT}: high;
- κ_{compat}: no single operator collapses order, interval, topology, dimension, product, theorem, scale, dynamics, and empirical readout;
- h_{obs}: high.

Post-diagnostic

The hard residual family survives mining and diagnostic audit:

$$D_{\text{hard}} = (D_{\mathcal{C}}^{\text{flat/global}}, D_{\text{car}}, D_d^{\text{wit}}, D_{\text{prod}}, D_{\text{thm}}, D_{\text{count}}, D_{\text{dens}}, D_{\text{anchor}}, D_{\text{metric}}, D_{\text{dyn}}, D_{\text{emp}}).$$

Classification

$$\boxed{\text{critical relation locus.}}$$

The proper closure is the target-presentation calibration atlas.

6A. Reader-replicable diagnostic rubric

The diagnostic table is useful only if another researcher can reproduce or challenge its rows.

A row is classified by the following procedure.

6A.1 Step 1 — Declare the target regime

The first operation is to narrow the target. A domain such as “Newtonian mechanics” or “quantum measurement” is not enough.

One must declare:

$$D'_Y = \text{the target distinctions being tested.}$$

For example:

- “Newtonian mechanics inside inertial-frame classical motion” is a bounded target.

- “Newtonian mechanics as all of physics” is not the same target.
- “Schrödinger dynamics in Hilbert space” is bounded.
- “Quantum measurement with definite classical outcomes” is a different target.

6A.2 Step 2 — Fill the target-demand matrix

For the declared target, mark which faces and target classes are active:

$$(\mathsf{Z}, \mathsf{R}, \mathsf{O})$$

across the target columns:

$$\prec, \mathcal{I}, \tau, \mathcal{C}, d, \mu/\text{scale}, \text{dyn}, \text{emp}.$$

A low-demand row activates few independent target classes. A high-demand row activates many independent classes.

6A.3 Step 3 — Estimate the six pre-diagnostic coordinates

For the row, assign qualitative values to:

$$\mathcal{D}_{\text{pre}} = (\rho_{\text{face}}, b_{\text{targ}}, s_{\text{sub}}, n_{\text{NFT}}, \kappa_{\text{compat}}, h_{\text{obs}}).$$

Use the following reproducible rubric:

Coordinate	Low	Moderate	High
ρ_{face}	one main target-face family	several linked faces	many independent faces
b_{targ}	one target presentation	few target presentations	many non-equivalent targets
s_{sub}	no role opens recursively	one role opens	several roles open recursively
n_{NFT}	few/no target jumps	some target jumps	many no-free-transfer jumps
κ_{compat}	strong known operator/law	partial operator/law	absent or contested
h_{obs}	simple readout/admission	nontrivial readout	difficult observer/readout/empirical access

The important point: high difficulty alone is not enough. Criticality requires high target demand **and** absent/insufficient terminal compatibility operator **and** persistent diagnostically diagnostically audited residuals.

6A.4 Step 4 — Apply decision rules

The table separates **primary classes** from **modifiers**.

Primary classes are:

$$extboundedbridge, \quad extfiniteunificationatlas, \quad extcriticalrelationlocus, \quad extaccidentalunderdeclaratio$$

Modifiers such as “high-valence,” “finite-to-critical,” “bounded-to-finite,” or “target-calibration problem” record a row’s location near a boundary between primary classes. They are not new primitive classes.

Bounded bridge

Classify as bounded when:

$$\rho_{\mathrm{face}}, b_{\mathrm{targ}}, s_{\mathrm{sub}}, n_{\mathrm{NFT}}$$

are low/moderate and:

$$\kappa_{\mathrm{compat}}$$

is strong for the declared target.

Finite unification atlas

Classify as finite unification when target demand is high but a compatibility operator/law collapses the branches into a finite admitted system.

Critical relation locus

Classify as critical only when:

1. target demand is high;
2. bridge-valence is high;
3. substitution depth is high;
4. no-free-transfer count is high;
5. no terminal compatibility operator collapses the target;
6. residuals persist after diagnostic audit.

Mixed / borderline / underdetermined

Return mixed or borderline when subtargets belong to different classes or when the classification margin is small.

Return underdetermined when source, target, observer slice, or compatibility operator is not declared enough.

6A.5 Step 5 — Check diagnostic error and margin

A row should not receive an admitted classification unless:

$$\Delta_{\mathrm{diag}} \preceq \varepsilon_{\mathrm{diag}}$$

and:

$$m_C > \eta_{\mathrm{class}}.$$

Otherwise the status column must say conditional, borderline, mixed, underdetermined, blocked, or diagnostic-pending.

6A.6 Worked row: harmonic oscillator

Declared target regime:

$$\text{given mass/spring system; solve motion in classical regime.}$$

Active target classes are narrow: motion inside one classical model. The equation:

$$m\ddot{x} + kx = 0$$

is a strong compatibility law. There is little substitution depth and few no-free-transfer jumps. Therefore the row is a bounded bridge.

If the target were changed to “recover quantum oscillator spectrum,” the classification would change.

6A.7 Worked row: Maxwellian electromagnetism

Declared target regime:

classical EM field/source/wave unification.

The target has many faces: electric/magnetic fields, sources, waves, conservation, and propagation. Before the field equations, the bridge-valence is high. Maxwell equations provide a compatibility operator that collapses the branch family into a finite unification atlas. Therefore the row is finite unification, not critical.

If the target includes QED, electroweak unification, or gravity, those are additional target regimes.

6A.8 Worked row: quantum measurement

Declared target regime:

bridge from quantum formalism to definite classical outcome/readout.

The target activates dynamics, probability, observer/readout, update, ontology/interpretation, and empirical outcome distinctions. There is no single universally admitted compatibility operator across target presentations. Different interpretations supply different bridges. Therefore the row is critical/mixed and target-dependent, not cleanly bounded.

6A.9 Worked row: observer-slice AC / continuum node

Declared target regime:

observer-slice AC presentation calibrated to multiple stronger targets.

The target activates order, intervals, topology, charting, carrier coherence, dimension, product policy, theorem applicability, counting, density, anchoring, metric scale, dynamics, and empirical readout.

No single terminal compatibility operator collapses all target faces. The hard residual family survives mining and diagnostic audit. Therefore this row is classified as a critical relation locus.

6B. Diagnostic landscape table

The following table is a schematic calibration landscape, not a historical reconstruction.

Its purpose is to show that the diagnostic does not classify everything as critical merely because a problem is difficult. It distributes cases across bounded bridges, finite unification atlases, critical loci, mixed cases, and underdetermined cases.

The classifications are admitted only relative to the **declared target regime** in the table. Changing the target regime can change the classification.

#	Case / domain	Declared target regime	Pre-diagnostic signature	Compatibility operator / bridge law	Predicted class	Diagnostic status	Main residual / guardrail
1	Classical harmonic oscillator	given mass/spring system; solve motion in classical regime	low face demand, low target-valence	equation of motion $m\ddot{x} + kx = 0$	bounded bridge	admitted schematic	regime boundary; no quantum/relativistic target
2	Kepler / Newtonian two-body problem	inverse-square two-body motion in Newtonian regime	moderate face demand, low target-valence	Newtonian force law + equations of motion	bounded bridge	admitted schematic	idealization boundary; perturbations external to target
3	Newtonian mechanics	inertial-frame classical motion for declared forces	moderate face demand, low-to-moderate target-valence	Newtonian laws + inertial-frame structure	bounded bridge	admitted schematic	not global physics; relativistic/quantum targets not included
4	Heat equation / diffusion	continuum-medium diffusion with declared initial/boundary data	moderate face demand, low target-valence	PDE + boundary/initial conditions	bounded bridge	conditional schematic	medium assumptions and boundary conditions
5	Maxwellian electromagnetism	classical EM field/source/wave unification	high face demand, high target-valence	Maxwell equations unify field/source/wave constraints	finite unification atlas	admitted schematic	does not include quantum electrodynamics or gravity
6	Special relativity	inertial-frame transformations with invariant light speed / interval	high face demand, finite target set	Lorentz transformations / Minkowski interval	finite unification atlas	admitted schematic	gravity and quantum targets excluded
7	General relativity	classical metric-gravity target with Einstein equations	high face demand, high target-valence	Einstein field equations + differential-geometric structure	finite unification atlas	conditional schematic	quantum, singularity, empirical model-building residuals excluded

#	Case / domain	Declared target regime	Pre-diagnostic signature	Compatibility operator / bridge law	Predicted class	Diagnostic status	Main residual / guardrail
8	Electroweak unification	electroweak gauge theory with symmetry breaking	high face demand, finite target set	gauge symmetry + symmetry breaking structure	finite unification atlas	conditional schematic	strong/gravity/dark-sector targets excluded
9	Standard Model gauge structure	QFT gauge-interaction target with declared field/representation content	high face demand, high target-valence	gauge symmetry + representations + renormalizable interactions	high-valence finite unification atlas	admitted schematic	gravity, hierarchy, dark matter, full neutrino sector residuals not closed by target
10	Renormalization group / EFT matching	scale-dependent effective theory matching inside declared cutoff/regime	moderate-to-high face demand	flow/matching equations and cutoff discipline	finite bridge atlas	conditional schematic	target-regime boundaries and matching conditions
11	Thermodynamics to statistical mechanics	equilibrium macro-behavior from microstate/statistical assumptions	high face demand, many bridge assumptions	ensemble/limit/typicality/ergodic-style bridges	mixed / conditional finite atlas	borderline	micro-macro bridge and equilibrium assumptions
12	Quantum Schrödinger dynamics	unitary evolution in declared Hilbert-space/Hamiltonian target	moderate face demand	unitary dynamics / Hamiltonian	bounded bridge inside target	admitted schematic	measurement/classi readout not included
13	Quantum measurement / classical outcome problem	bridge from quantum formalism to definite outcome/readout	high face demand, high bridge-valence	no single universally admitted compatibility operator	critical or mixed locus	borderline / programme-dependent	observer/readout, ontology, probability, update residuals
14	Black-hole thermodynamics / holographic reconstruction	entropy/area/bulk-boundary reconstruction target	high face demand, high target-valence	partial compatibility via area/entropy/boundary-style bridges	critical / finite-atlas / borderline	mixed	dictionary, dynamics, bulk reconstruction, empirical target

#	Case / domain	Declared target regime	Pre-diagnostic signature	Compatibility operator / bridge law	Predicted class	Diagnostic status	Main residual / guardrail
15	Causal-set faithful embedding / manifold-likeness	locally finite order approximating manifold-like continuum	high face demand, high target-valence	sprinkling/faithfulness conditions if supplied	critical or conditional branch atlas	conditional	density, dimension, topology, manifold approximation
16	Observer-slice AC / continuum node	observer-slice AC presentation calibrated to multiple stronger targets	high face demand, high bridge-valence, deep substitution	no terminal operator for all target faces	critical relation locus	admitted by post-diagnostic for current task	hard residual family D_{hard}
17	Quantum gravity / spacetime emergence broadly construed	recover spacetime-like target from non-spatiotemporal formalism	very high face demand and target-valence	no single accepted compatibility operator	critical relation locus	underdetermined / mixed	depends strongly on target presentation
18	Empirical physical spacetime recovery from a formal structure	empirical/measurement-calibrated spacetime role recovery	high face demand, high observer-readout hardness	requires empirical-readout / measurement bridge	critical / target-calibration problem	conditional	Δ_{emp}, Δ_{dyn}, Δ_{metric}
19	Rigid-body mechanics	constrained finite-dimensional classical mechanics	moderate face demand, moderate target-valence	Euler equations + constraint forces / symplectic or variational structure	bounded-to-finite bridge atlas	conditional schematic	constraints and dissipation require declared target
20	Classical optics from wave/eikonal approximation	geometric-optics rays from short-wavelength wave regime	moderate face demand, low-to-moderate target-valence	eikonal/geometric-optics approximation bridge	bounded bridge	conditional schematic	approximation boundary; diffraction/quantum targets excluded

#	Case / domain	Declared target regime	Pre-diagnostic signature	Compatibility operator / bridge law	Predicted class	Diagnostic status	Main residual / guardrail
21	Lagrangian/Hamiltonian equivalence	regular mechanical systems with non-singular Legendre transform	moderate face demand, finite target set	Legendre transform under regularity	bounded bridge	conditional schematic	singular systems and constraints create extra branches
22	Noether theorem	conservation law from symmetry in declared variational setting	moderate face demand, finite target set	symmetry-action-conservation bridge	finite bridge theorem	admitted schematic	boundary terms and variation class must be declared
23	Navier-Stokes local well-posedness	local-in-time PDE solution in declared function spaces	moderate face demand, low target-valence	PDE estimates in declared function spaces	bounded bridge	conditional schematic	global regularity target is different
24	Navier-Stokes global regularity	global smoothness/no-blowup target	high face demand, high observer/readout hardness	no single diagnostic operator supplied at this level	critical / underdetermined	diagnostic pending	scale, blowup, function-space, global boundary residuals
25	Turbulence closure / high-Reynolds prediction	reduced/statistical closure for turbulent flow prediction	high face demand, high target-valence	modelling closures, statistical bridges	critical relation locus	mixed / conditional	scale cascade, empirical closure, stochastic readout residuals
26	Ising model phase transition	lattice/statistical target with thermodynamic-limit or exact/asymptotic behavior	moderate face demand, finite target set	partition function / renormalization / exact or asymptotic machinery	finite unification atlas	conditional schematic	depends on dimension, thermodynamic limit, universality target
27	Critical phenomena / universality classes	classify cross-system scaling behavior near criticality	high face demand, high target-valence	RG fixed points and universality maps	finite-to-critical atlas	mixed	universality bridge and relevant operators must be declared

#	Case / domain	Declared target regime	Pre-diagnostic signature	Compatibility operator / bridge law	Predicted class	Diagnostic status	Main residual / guardrail
28	Bayesian inference	posterior inference in declared statistical model	low-to-moderate face demand	Bayes rule + likelihood/prior/model class	bounded bridge	admitted schematic	model misspecification and causal target not included
29	Causal inference from observational data	infer intervention/counterfactual claims from observational/statistical data	high face demand, factual-type jumps	causal graph / intervention assumptions	conditional bridge atlas	conditional / borderline	no-free-transfer from association to intervention/counterfactual
30	Machine-learning prediction	fixed-distribution supervised task with declared loss	moderate face demand, low target-valence	empirical risk / generalization assumptions	bounded bridge	conditional schematic	distribution shift and causal explanation not included
31	OOD generalization / robust agency	preserve performance/reasoning under environment/task shift	high face demand, high target-valence	invariance/causal/robustness assumptions, no single universal bridge	critical / mixed	underdetermined	environment shift, task drift, evaluator/readout residuals
32	Biological evolution by natural selection	population-genetic model with variation, inheritance, selection dynamics	high face demand but finite target within model	fitness, inheritance, variation, selection dynamics	finite unification atlas	conditional schematic	open-ended evolution and intelligence targets not included
33	Abiogenesis / origin-of-life transition	transition from chemistry to self-maintaining/evolving life-like system	very high face demand, many target types	partial chemistry, replication, metabolism, compartment bridges	critical relation locus	underdetermined / mixed	boundary between chemistry, information, metabolism, environment
34	Consciousness / subjective experience from neural dynamics	bridge neural/functional structure to phenomenal/report/readout target	very high face demand, high observer/readout hardness	no single admitted bridge across target presentations	critical relation locus	underdetermined	first-person/third-person readout, identity, function, report residuals

#	Case / domain	Declared target regime	Pre-diagnostic signature	Compatibility operator / bridge law	Predicted class	Diagnostic status	Main residual / guardrail
35	Measurement problem in quantum foundations	account for quantum-to-classical outcome/update/probability target	high face demand, high target-valence	interpretation-specific bridge laws	critical / mixed	mixed / target-dependent	ontology, probability, update, observer/readout residuals
36	Formal theory to empirical science bridge	calibrate formal structure to empirical measurement/intervention target	high face demand, high observer/readout hardness	measurement model + calibration + intervention/validation bridges	critical relation locus	conditional / mixed	empirical readout, operationalization, instrument, statistics residuals

6B.1 What the table shows

The table has all major outputs:

$$\boxed{\text{bounded bridge}}$$

$$\boxed{\text{finite unification atlas}}$$

$$\boxed{\text{critical relation locus}}$$

$$\boxed{\text{mixed / borderline / underdetermined}}$$

This distribution is important. It shows the diagnostic does not label every hard or important problem as critical.

6B.2 Why thirty-six cases

Thirty-six cases is a useful upper bound for this version of the diagnostic table. It is large enough to show a fuller landscape across physics, mathematics, statistics, biology, cognitive science, and empirical-bridge problems, but still compact enough to avoid turning the paper into a survey.

The table spans:

1. low-valence bounded classical targets;
2. finite unification targets;
3. high-valence but compatibility-stabilized targets;
4. statistical, causal, and learning targets;
5. biological and cognitive transition targets;
6. micro-macro and measurement borderline cases;
7. spacetime/continuum and empirical-bridge critical-locus cases.

6B.3 How to use the table

Each row is a diagnostic hypothesis. A researcher may refine a row by:

1. narrowing the target regime;
2. changing the observer-slice assumptions;
3. adding a compatibility operator;
4. splitting mixed cases into subtargets;
5. increasing or decreasing Δ_{diag}.

Thus the table is replicable and revisable, not a fixed historical verdict.

6C. Diagnostic trace table: making the 36 cases reproducible

The 36-case landscape table is not the diagnostic audit. It is a **compact diagnostic trace table**.

A full paper-construction audit is a different internal workflow. The table below is a smaller row-level diagnostic audit. It records enough information for a reader to reconstruct the classification.

Use the shorthand:

$$L = \text{low}, \quad M = \text{moderate}, \quad H = \text{high}, \quad VH = \text{very high}.$$

For compatibility operator availability:

$$S = \text{strong}, \quad P = \text{partial}, \quad N = \text{none/contested}.$$

The coordinate vector is:

$$(\rho, b, s, n, \kappa, h) = (\rho_{\text{face}}, b_{\text{targ}}, s_{\text{sub}}, n_{\text{NFT}}, \kappa_{\text{compat}}, h_{\text{obs}}).$$

#	Case	Declared target regime shorthand	Coordinate trace $(\rho, b, s, n, \kappa, h)$	Active theorem/no-go subtree	Class/status
1	Harmonic oscillator	classical mass-spring motion	(L, L, L, L, S, L)	bounded equation law / local boundary	bounded / admitted schematic
2	Kepler two-body	Newtonian inverse-square two-body motion	(M, L, L, L, S, L)	Newtonian dynamics / idealization boundary	bounded / admitted schematic
3	Newtonian mechanics	inertial-frame classical motion	(M, M, L, L, S, M)	inertial-frame boundary / dynamics law	bounded / admitted schematic
4	Heat equation	declared continuum diffusion problem	(M, L, L, L, S, M)	PDE + boundary/initial data	bounded / conditional schematic
5	Maxwell EM	classical EM unification	(H, H, M, M, S, M)	compatibility law / finite field-equation atlas	finite unification / admitted schematic

#	Case	Declared target regime shorthand	Coordinate trace $(\rho, b, s, n, \kappa, h)$	Active theorem/no-go subtree	Class/status
6	Special relativity	inertial-frame Lorentz target	(H, M, M, M, S, M)	Lorentz compatibility / interval invariant	finite unification / admitted schematic
7	General relativity	classical metric-gravity target	(H, H, H, H, S, H)	differential-geometry + field equation bridge	finite unification / conditional schematic
8	Electroweak	gauge theory with symmetry breaking	(H, H, H, H, S, H)	gauge symmetry / representation / breaking	finite unification / conditional schematic
9	Standard Model	QFT gauge-structure target	(H, H, H, H, S, H)	gauge/representation compatibility	high-valence finite unification / admitted schematic
10	RG / EFT	scale-dependent matching target	(M, H, M, M, P, M)	scale bridge / matching / cutoff discipline	finite bridge atlas / conditional schematic
11	Thermo to stat mech	equilibrium macro from micro-statistical target	(H, H, H, H, P, H)	micro-macro / limit / typicality subtrees	mixed finite atlas / borderline
12	Schrödinger dynamics	unitary Hilbert-space evolution	(M, L, L, L, S, M)	Hamiltonian/unitary dynamics	bounded inside target / admitted schematic
13	Quantum measurement	definite outcome/readout target	(H, H, H, H, N, VH)	observer/readout / update / ontology no-free-transfer	critical or mixed / borderline
14	Black-hole thermodynamics / holography	entropy/bulk-boundary reconstruction target	(VH, H, H, H, P, H)	dictionary / bulk reconstruction / thermodynamic bridge	critical-finite borderline / mixed
15	Causal-set faithful embedding	manifoldlike approximation from locally finite order	(H, H, H, H, P, H)	density / dimension / topology / embedding no-gos	critical or conditional atlas / conditional
16	Observer-slice AC node	calibration to multiple stronger targets	(VH, VH, H, VH, N, H)	full target-atlas / programme theorem/no-go / residual persistence	critical / admitted by post-diagnostic

#	Case	Declared target regime shorthand	Coordinate trace $(\rho, b, s, n, \kappa, h)$	Active theorem/no-go subtree	Class/status
17	Quantum gravity emergence	spacetime-like recovery from non-spatiotemporal formalism	(VH, VH, VH, VH, N, VH)	target-presentation calibration / spacetime emergence	critical / underdetermined / mixed
18	Empirical physical spacetime	empirical measurement-calibrated spacetime role	(VH, VH, H, VH, B, VH)	empirical readout / instrument / validation bridge	critical target-calibration / conditional
19	Rigid-body mechanics	constrained classical mechanics	(M, M, M, M, S, M)	constraint / variational / symplectic machinery	bounded-to-finite atlas / conditional
20	Classical optics	geometric optics from wave approximation	(M, M, M, M, P, M)	approximation / eikonal bridge	bounded / conditional
21	Lagrangian/Hamiltonian equivalence	regular systems under Legendre transform	(M, M, M, M, S, M)	compatibility transform / singular-system no-go	bounded / conditional
22	Noether theorem	conservation from variational symmetry	(M, M, M, M, S, M)	symmetry-action-conservation bridge	finite bridge theorem / admitted schematic
23	Navier-Stokes local	local well-posedness in function space	(M, L, M, M, P, M)	PDE estimates / function-space boundary	bounded / conditional
24	Navier-Stokes global	global regularity/no blowup target	(H, H, H, H, N, H)	scale/blowup/global-boundary residual	critical or underdetermined / diagnostic pending
25	Turbulence closure	reduced/statistical turbulent-flow prediction	(VH, H, H, H, R, VH)	scale cascade / empirical closure / stochastic readout	critical / mixed
26	Ising phase transition	lattice/statistical thermodynamic target	(M, H, M, M, S, M)	partition function / RG / thermodynamic limit	finite unification / conditional
27	Critical phenomena	universality class / scaling target	$(H, H, H, H, S/R, H)$	RG fixed point / universality bridge	finite-to-critical atlas / mixed
28	Bayesian inference	posterior inference in declared model	(M, L, L, L, S, M)	Bayes rule / model-class boundary	bounded / admitted schematic

#	Case	Declared target regime shorthand	Coordinate trace $(\rho, b, s, n, \kappa, h)$	Active theorem/no-go subtree	Class/status
29	Causal inference	intervention/counterfactual from observation	(H, H, H, H, P, H)	association-to-intervention no-free-transfer	conditional bridge atlas / borderline
30	ML prediction	fixed-distribution supervised loss target	(M, L, M, M, P, M)	generalization/loss/evaluator bridge	bounded / conditional
31	OOD generalization / agency	environment-shift robust performance target	(H, H, H, H, N, H)	invariance / causal / evaluator drift	critical or mixed / underdetermined
32	Evolution by natural selection	population-genetic variation/inheritance/selection target	$(H, M, H, M, S/P, M)$	fitness/inheritance/selection dynamics	finite unification / conditional
33	Abiogenesis	chemistry-to-life transition target	(VH, VH, VH, VH, N, VH)	metabolism/replication/compartment boundary	critical / underdetermined
34	Consciousness	neural-to-phenomenal/report target	(VH, VH, VH, VH, N, VH)	first-person/third-person readout / identity bridge	critical / underdetermined
35	Quantum measurement detailed	outcome/update/probability target	$(H, H, H, H, P/N, VH)$	interpretation-specific bridge / observer update	critical or mixed / target-dependent
36	Formal theory to empirical science	empirical measurement/intervention calibration target	(H, H, H, H, P, VH)	measurement model / operationalization / validation	critical / conditional / mixed

6C.1 How to read a trace row

A row should be reproducible as follows.

First, read the declared target regime. Then estimate each coordinate using the rubric in §6A. If the compatibility operator is strong and the target demand is low, the row tends toward bounded bridge. If target demand is high but compatibility is strong, the row tends toward finite unification atlas. If target demand is high, compatibility is absent or contested, substitution is deep, no-free-transfer count is high, and observer/readout hardness is high, the row tends toward critical locus.

The active theorem/no-go subtree explains which programme machinery justifies the row.

6C.2 Example of reconstructing a row

For row 16, the declared target is not merely “continuum.” It is:

observer-slice AC presentation calibrated to multiple stronger targets.

That target activates order, intervals, topology, charting, carrier coherence, dimension, product policy, theorem applicability, count, density, anchoring, metric scale, dynamics, and empirical readout.

Thus:

$$(\rho, b, s, n, \kappa, h) = (VH, VH, H, VH, N, H).$$

The active subtree is the full target-atlas and diagnostic diagnostically audited residual-persistence machinery. Since no terminal compatibility operator collapses all target faces and the residuals persist after diagnostic audit, the row is classified as a critical relation locus.

6C.3 What this table is not

This is not a full paper-construction audit. It is a diagnostic trace summary.

A full paper-construction audit would be an internal workflow exercise. The present table is a problem diagnostic: it records target regime, coordinate trace, active theorem/no-go subtree, classification, status, and residual/guardrail.

This table gives the minimum row-level trace needed for reader reproducibility.

6C.4 Row-level action implications

The diagnostic landscape table should not stop at classification. Each row also has an action implication.

Use the following class-to-action rule:

Classification type	Default action implication
bounded bridge	close / verify within declared target; do not over-atlas
finite unification atlas	identify the compatibility operator and its residual boundary
high-valence finite unification atlas	preserve the compatibility operator and map external residuals
conditional bridge / atlas	declare hypotheses and calibrator; test stability under target changes
mixed / borderline	split target presentations and rerun the diagnostic
underdetermined / diagnostic-pending	declare missing target, readout, compatibility, or data
critical relation locus	run AtlasMap + CompatSearch

6C.4.1 Action table for the 36 cases

#	Case	Classification	Row-level action implication
1	Classical harmonic oscillator	bounded bridge	close/verify inside classical mass-spring target; no atlas expansion needed
2	Kepler / Newtonian two-body	bounded bridge	close/verify within idealized inverse-square regime; record perturbation boundary

#	Case	Classification	Row-level action implication
3	Newtonian mechanics	bounded bridge	close within inertial-frame classical target; do not promote to all physics
4	Heat equation / diffusion	bounded bridge / conditional	declare medium and boundary conditions; verify PDE target
5	Maxwellian electromagnetism	finite unification atlas	identify Maxwell equations as κ_{compat}; map QED/gravity residuals outside target
6	Special relativity	finite unification atlas	identify Lorentz/Minkowski compatibility; record gravity/quantum exclusions
7	General relativity	finite unification atlas / conditional	identify Einstein-equation compatibility; separate quantum/singularity/empirical residuals
8	Electroweak unification	finite unification atlas / conditional	identify gauge/breaking compatibility; record strong/gravity/dark residuals
9	Standard Model gauge structure	high-valence finite unification atlas	preserve gauge/representation compatibility; map unresolved external residuals
10	RG / EFT matching	finite bridge atlas / conditional	declare cutoff/matching calibrator; map regime boundaries
11	Thermodynamics to statistical mechanics	mixed / conditional finite atlas	split equilibrium, limit, typicality, and micro-macro target presentations
12	Schrödinger dynamics	bounded inside target	close unitary Hilbert-space dynamics; separate measurement/readout problem
13	Quantum measurement	critical or mixed	run AtlasMap over interpretations and CompatSearch over outcome/update operators

#	Case	Classification	Row-level action implication
14	Black-hole thermodynamics / holography	critical / finite-atlas borderline	split entropy, dictionary, reconstruction, dynamics, empirical targets
15	Causal-set faithful embedding	critical or conditional atlas	split embedding/manifoldlikeness/Hauptver branches; search density-closeness operator
16	Observer-slice AC node	critical relation locus	run full target-presentation atlas and compatibility-operator search
17	Quantum gravity / spacetime emergence	critical / underdetermined	declare target presentation before classification; then atlas-map candidate programmes
18	Empirical physical spacetime recovery	critical / conditional	declare empirical-readout calibrator and map model-to-measurement bridge
19	Rigid-body mechanics	bounded-to-finite atlas / conditional	declare constraints; split regular, constrained, dissipative variants
20	Classical optics	bounded / conditional	declare approximation regime; record diffraction/quantum residuals
21	Lagrangian/Hamiltonian equivalence	bounded / conditional	verify regularity; split singular/constraint cases
22	Noether theorem	finite bridge theorem	declare action, symmetry, boundary terms, current readout
23	Navier-Stokes local	bounded / conditional	declare function spaces and local time interval; separate global target
24	Navier-Stokes global	critical / underdetermined	atlas-map blowup/regularity regimes; search global a priori compatibility estimate
25	Turbulence closure	critical / mixed	map closure models and scale cascades; search stable reduced compatibility law

#	Case	Classification	Row-level action implication
26	Ising phase transition	finite unification atlas / conditional	declare dimension/limit regime; identify partition/RG compatibility
27	Critical phenomena	finite-to-critical atlas / mixed	split universality classes; search RG fixed-point compatibility per class
28	Bayesian inference	bounded bridge	close under declared prior/likelihood/model class; record misspecification boundary
29	Causal inference	conditional bridge atlas	declare graph/intervention assumptions; block association-to-causation transfer
30	ML prediction	bounded / conditional	close under fixed distribution/loss; record distribution-shift boundary
31	OOD generalization / robust agency	critical / mixed	atlas-map environment/task shifts; search invariance/causal compatibility operators
32	Evolution by natural selection	finite unification atlas / conditional	declare population-genetic model; separate open-ended/intelligence targets
33	Abiogenesis	critical / underdetermined	atlas-map metabolism/replication/compar targets; search transition compatibility
34	Consciousness	critical / underdetermined	atlas-map functional, neural, phenomenal, report targets; search readout/identity compatibility
35	Quantum measurement detailed	critical / mixed	split interpretation-specific bridges; test out-come/update/probability compatibility
36	Formal theory to empirical science	critical / conditional-mixed	declare measure-ment/instrument/statistical calibrator; map empirical bridge atlas

6C.4.2 What this patch changes

The diagnostic table now has three layers:

$$\boxed{\text{classification} \to \text{status} \to \text{action implication.}}$$

Thus the table is not only descriptive. It is method-directing.

6D. Scoring-sheet provenance and calibrator status

The scoring sheet must be distinguished from the diagnostic coordinates.

The coordinates:

$$(\rho_{\text{face}}, b_{\text{targ}}, s_{\text{sub}}, n_{\text{NFT}}, \kappa_{\text{compat}}, h_{\text{obs}})$$

are derived from relation-first bridge-audit machinery.

The numerical coding:

$$L = 0, \quad M = 1, \quad H = 2, \quad VH = 3$$

is not uniquely derived. It is a **declared diagnostic calibrator**.

Definition 6D.1 — Ordinal diagnostic calibrator

The ordinal diagnostic calibrator is:

$$\boxed{\mathfrak{K}^{\text{ord}}_{\text{diag}} = (D_{\text{diag}}, V_{\text{ord}}, \preceq_{\text{ord}}, \mathcal{E}_{\text{diag}}, \text{Adm}_{\text{diag}}, \text{Update}_{\text{diag}}).}$$

Where:

1. D_{diag} is the diagnostic target: classify the target-bridge problem;
2. $V_{\text{ord}} = \{0, 1, 2, 3\}$ is the ordinal value set;
3. $\preceq_{\text{ord}}$ is the order $0 < 1 < 2 < 3$;
4. $\mathcal{E}_{\text{diag}}$ records diagnostic error and margin;
5. Adm_{diag} admits or rejects a classification;
6. $\text{Update}_{\text{diag}}$ updates the row if the target regime, coordinate value, or active theorem/no-go subtree is challenged.

Definition 6D.2 — Minimal ordinal readout

The coding:

$$L = 0, \quad M = 1, \quad H = 2, \quad VH = 3$$

is a minimal ordinal readout with four levels:

Level	Meaning
L	low / narrow target demand
M	moderate / several linked demands
H	high / many independent demands
VH	very high / many independent demands plus deep branching

The four-level scale is chosen because the diagnostic needs to distinguish:

1. bounded;
2. moderate/conditional;
3. high-valence finite unification;
4. critical / very high branching.

A binary scale would be too coarse. A much finer scale would invite false precision.

Theorem 6D.3 — Scoring Sheet Status Theorem

The ordinal scoring sheet is not a theorem-unique consequence of RFOP. It is a declared calibrator/readout required for reproducible diagnostic classification.

Proof

The programme derives the diagnostic coordinates from bridge-audit machinery. But assigning a numerical or ordinal value to a coordinate is a readout operation. By No Undeclared Observer, any readout must be declared. By Recursive Bridge Calibration, a repeatable admission judgment requires a calibrator: value domain, order, evaluator, admission rule, and update rule. Therefore a scoring sheet is required if row classifications are to be reproducible. However, relation-first machinery does not uniquely select the four-level ordinal coding. Other declared calibrators may refine or coarsen it. Thus the scoring sheet is programme-native as a declared calibrator, but not unique as a theorem. □

Corollary 6D.4 — Monotone refinements are admissible

A different scoring sheet is admissible if it is declared and monotone with respect to diagnostic demand.

For example, a five-level or weighted scoring system may be used if it preserves the target-demand ordering and states its own thresholds.

Corollary 6D.5 — Table classifications are calibrator-relative

A table classification is admitted relative to:

$$\mathfrak{K}_{\mathrm{diag}}^{\mathrm{ord}}.$$

If a reader changes the scoring calibrator, the row may need to be recomputed.

This is not a contradiction. It is the diagnostic version of local exactness: classification is exact only relative to the declared diagnostic calibrator.

6E. General form: granularity-indexed diagnostic calibrator family

The previous section distinguished the diagnostic coordinates from the four-level ordinal scoring sheet. This section gives the general programme-native form of the scoring sheet.

The key point is:

$$\boxed{\text{the existence and form of a scoring calibrator are programme-native;}}$$

but:

$$\boxed{\text{the exact granularity of the score chart is declared.}}$$

6E.1 Source in the programme papers

The general form is forced by four pieces of existing programme machinery.

Programme machinery	Contribution to scoring form
Triadic closure / substitution / typed micro-triads	target demand decomposes by $\mathsf{Z}, \mathsf{R}, \mathsf{O}$, face-depth, subconfiguration failure, and role-recursion depth
Charted Organization	diagnostic scores are charted readouts over task-visible distinctions; different granularities are different charts
Recursive Chart Learning	score use requires evaluator, admissible region, calibration domain, and update rule
Recursive Bridge Calibration	score sheets are calibrators bounded between underdeclaration and terminal totalization

Thus the score sheet is a charted calibrator over a diagnostic relation-presentation.

Definition 6E.2 — Diagnostic complexity presentation

For a bridge problem (X, D'_Y), define the **diagnostic complexity presentation**:

$$\mathfrak{P}_{\text{diag}}(X, D'_Y) = (F_{\text{face}}, F_{\text{targ}}, F_{\text{sub}}, F_{\text{NFT}}, F_{\text{compat}}, F_{\text{obs}}).$$

These are not numerical primitives. They are relation-first readout domains:

Component	Meaning
F_{face}	face-demand configurations induced by $\mathsf{Z}, \mathsf{R}, \mathsf{O}$ requirements
F_{targ}	target-presentation branch family
F_{sub}	recursive role-substitution depth patterns
F_{NFT}	no-free-transfer/type-promotion obligations
F_{compat}	compatibility-operator / face-recovery availability
F_{obs}	observer/readout/evaluator/calibrator burden

The pre-diagnostic coordinates are charted readouts of these domains.

Definition 6E.3 — Diagnostic granularity

A **diagnostic granularity** is a tuple:

$$\Gamma = (\pi_\rho, \pi_b, \pi_s, \pi_n, \pi_\kappa, \pi_h),$$

where each π is a finite ordered partition or finite ordered quotient of the corresponding diagnostic complexity domain.

For example:

$$\pi_\rho : F_{\text{face}} \to V_\rho^\Gamma.$$

The target value domains:

$$V_\rho^\Gamma, V_b^\Gamma, V_s^\Gamma, V_n^\Gamma, V_\kappa^\Gamma, V_h^\Gamma$$

are finite ordered chains or finite ordered sets.

Definition 6E.4 — Granularity-indexed diagnostic score chart

Given Γ, the diagnostic score chart is:

$$\boxed{S_\Gamma : \mathfrak{P}_{\mathrm{diag}}(X, D'_Y) \to V_\rho^\Gamma \times V_b^\Gamma \times V_s^\Gamma \times V_n^\Gamma \times V_\kappa^\Gamma \times V_h^\Gamma.}$$

It is a charted readout of diagnostic complexity.

The four-level score sheet used in the table is the special case:

$$V_\rho^{\Gamma_4} = V_b^{\Gamma_4} = V_s^{\Gamma_4} = V_n^{\Gamma_4} = V_h^{\Gamma_4} = \{L < M < H < VH\}$$

and:

$$V_\kappa^{\Gamma_4} = \{S < P < N\}$$

with obstruction coding $S \mapsto 0$, $P \mapsto 1$, $N \mapsto 2$.

Definition 6E.5 — Granularity refinement

A granularity Γ' refines Γ, written:

$$\Gamma \preceq \Gamma',$$

when there is a monotone coarsening map:

$$q_{\Gamma' \to \Gamma} : V^{\Gamma'} \to V^\Gamma$$

such that:

$$S_\Gamma = q_{\Gamma' \to \Gamma} \circ S_{\Gamma'}.$$

Thus a finer scoring sheet may split H into several sublevels, but it must collapse monotonically back to the coarse chart.

Definition 6E.6 — Classification stability under granularity

A classification is **granularity-stable** over a refinement family $\mathcal{G}$ when all sufficiently fine declared granularities in $\mathcal{G}$ return the same primary class.

If the class changes under reasonable refinements, the proper status is:

$$\boxed{\text{borderline / granularity-sensitive.}}$$

Theorem 6E.7 — Granularity-Indexed Scoring Calibrator Theorem

For every finite declared target-demand problem (X, D'_Y), and every finite declared diagnostic granularity Γ, there exists a charted diagnostic calibrator:

$$\boxed{\mathfrak{K}_{\mathrm{diag}}^\Gamma}$$

whose score chart is:

$$S_\Gamma : \mathfrak{P}_{\text{diag}}(X, D'_Y) \to V^\Gamma.$$

The four-level scoring sheet is one admissible coarse member of this calibrator family, not the unique scoring theorem.

Proof

A finite declared target-demand problem has finitely many declared target distinctions, theorem/no-go checks, face-demand obligations, and active diagnostic domains. Triadic closure and substitution supply the face and role-recursion domains. No-Free-Transfer supplies the type-jump domain. Charted Organization says that any readout of these domains must be a charted relation-map. Recursive Chart Learning says that if the readout is used to evaluate or update a branch, the evaluator, admissible region, calibration domain, and update rule must be declared. Recursive Bridge Calibration says that repeatable admission requires a calibrator.

A finite granularity Γ is precisely such a declared finite chart of the diagnostic domains. Therefore it induces a score chart S_Γ and a diagnostic calibrator $\mathfrak{K}^\Gamma_{\text{diag}}$. Since many finite ordered partitions are possible, the four-level sheet is not unique. It is an admissible coarse chart. □

Corollary 6E.8 — Four-level scoring is a coarse chart, not an axiom

The table's L, M, H, VH scoring is a coarse observer-chart of diagnostic complexity.

It should not be read as primitive, unique, or universal.

Corollary 6E.9 — Finer diagnostic charts are allowed

A future version of the diagnostic may refine the score chart, for example:

$$\{L < M < H < VH\} \rightsquigarrow \{0, 1, 2, 3, 4, 5, 6\}$$

or to a weighted/lexicographic scoring domain.

Such a refinement is admissible if it is declared and monotone relative to the diagnostic-demand ordering.

Corollary 6E.10 — Table rows may update under refined granularity

If a finer granularity changes a row's classification, the result is not inconsistency. It means the row was coarse-chart sensitive.

The correct update is:

$$\text{Status} = \text{granularity-sensitive}$$

unless a refined diagnostic margin stabilizes the classification.

Register note

This is the exact analogue of charted organization: a chart makes distinctions visible at a declared resolution. A diagnostic score sheet makes bridge-complexity distinctions visible at a declared diagnostic resolution.

6F. Recursive triadic topology behind the score charts

The preceding section defined diagnostic score charts by granularity. This section identifies the underlying topology those charts read.

The safe claim is not:

$$\boxed{\text{the diagnostic topology is literally fractal.}}$$

The safe claim is:

$$\boxed{\text{the diagnostic topology is recursive triadic / closure-atlasic.}}$$

It may admit fractal-like realizations only after scale, refinement, and self-similarity data are declared.

Definition 6F.1 — Recursive triadic refinement step

Let:

$$T = (\mathsf{Z}_T, \mathsf{R}_T, \mathsf{O}_T)$$

be a declared triad in a diagnostic bridge problem.

A **recursive triadic refinement step** occurs when a nontrivial role-face or role-substitute:

$$\mathsf{X}_T \in \{\mathsf{Z}_T, \mathsf{R}_T, \mathsf{O}_T\}$$

is opened into a subordinate declared triad:

$$\mathsf{X}_T \Rightarrow (\mathsf{Z}_\mathsf{X}, \mathsf{R}_\mathsf{X}, \mathsf{O}_\mathsf{X}).$$

The step is admissible only if the subordinate triad lands in the parent closure context and does not leave a dangling role interface.

Definition 6F.2 — Recursive triadic topology

For a bridge problem (X, D'_Y), the **recursive triadic topology** is the depth-indexed closure-atlas structure:

$$\boxed{\mathfrak{T}_{\text{diag}}(X, D'_Y) = (\mathfrak{T}_0, \mathfrak{T}_1, \mathfrak{T}_2, \ldots; \mathrm{Sub}, \mathrm{Coup}, \Omega, \mathrm{Adm}, \mathrm{Update})}$$

where:

1. $\mathfrak{T}_0$ is the initial source–bridge–target triadic presentation;
2. $\mathfrak{T}_{n+1}$ is obtained from $\mathfrak{T}_n$ by admissible role-substitution, coupling, residual, trace, or readout refinement;
3. Sub records role-substitution steps;
4. Coup records landed coupling interfaces;
5. Ω records residuals;
6. Adm admits, blocks, or residualizes refinements;
7. Update updates the later branch field.

This is a topology in the programme sense: a charted closure structure of admissible local neighborhoods, refinements, transitions, and residual boundaries.

Definition 6F.3 — Diagnostic open neighborhoods

A **diagnostic open neighborhood** of a bridge problem is a finite family of refinements that preserve the same admitted classification under a declared granularity:

$$\mathcal{U}_\Gamma(X, D'_Y, C) = \{(X', D'_Y) : \mathrm{Diag}_\Gamma(X', D'_Y) \text{ returns class } C\}.$$

These neighborhoods are chart-relative. They depend on the diagnostic granularity Γ.

Definition 6F.4 — Coarse quotient of recursive triadic topology

A diagnostic score chart:

$$S_\Gamma : \mathfrak{P}_{\mathrm{diag}}(X, D'_Y) \to V^\Gamma$$

is a **coarse quotient** of $\mathfrak{T}_{\mathrm{diag}}(X, D'_Y)$ when it identifies all refinements whose complexity readouts lie in the same granularity class.

Thus the table's four-level sheet is a quotient:

$$\mathfrak{T}_{\mathrm{diag}} \to \{L < M < H < VH\}$$

for each demand coordinate.

Theorem 6F.5 — Score Charts Are Quotients of Recursive Triadic Topology

Every finite diagnostic granularity Γ defines a charted quotient of the recursive triadic topology:

$$\boxed{\mathfrak{T}_{\mathrm{diag}}(X, D'_Y) \longrightarrow S_\Gamma(X, D'_Y).}$$

Proof

The recursive triadic topology records the relation-first structure generated by target faces, role-substitution depth, no-free-transfer jumps, compatibility operators, observer/readout burden, residuals, couplings, and updates. A diagnostic granularity Γ partitions each of these readout domains into finite ordered classes. Applying these partitions to the topology identifies refinements that are indistinguishable at the declared diagnostic resolution. This is exactly a charted quotient/readout of the recursive triadic topology. □

Corollary 6F.6 — The four-level score sheet is a coarse topological chart

The table's L, M, H, VH score sheet is a coarse chart over:

$$\mathfrak{T}_{\mathrm{diag}}(X, D'_Y).$$

It is not primitive. It is not arbitrary. It is a declared finite quotient of recursive triadic closure structure.

Definition 6F.7 — Fractal-like triadic realization

A recursive triadic topology admits a **fractal-like realization** only when additional data are declared:

1. a scale or depth function:
$$\ell : \mathfrak{T}_n \to \Lambda;$$
2. a refinement law relating levels:
$$\mathfrak{T}_n \to \mathfrak{T}_{n+1};$$

3. a self-similarity or recurrence condition across levels;
4. a measure, dimension, or complexity readout stable enough to compare levels.

Without these data, the topology is recursive and closure-atlasic, but not literally fractal.

Theorem 6F.8 — Fractal Guardrail Theorem

Recursive triadic substitution alone does not imply fractal topology. It implies recursive triadic topology. A fractal-like interpretation is admissible only after a scale/readout/self-similarity calibrator is declared.

Proof

Recursive triadic substitution supplies repeated role-opening and closure-atlas refinement. A fractal claim requires more: a scale or depth comparison, a recurrence or self-similarity condition, and a measure/dimension/readout over refinements. These are not supplied by substitution alone. By No Undeclared Observer and Recursive Bridge Calibration, such readout and comparison data must be declared before the fractal-like claim is admissible. Therefore recursive triadic topology is programme-native, while fractal-like realization requires an additional declared calibrator. □

Corollary 6F.9 — Diagnostic granularity as depth-limited observation

A diagnostic granularity Γ is a depth-limited observer-chart of the recursive triadic topology. Coarser granularities see fewer refinement distinctions; finer granularities see more.

Thus the scoring sheet is naturally related to the programme's charted topology:

$$\boxed{\text{score sheet} = \text{observer-chart over recursive triadic complexity.}}$$

Register note

The paper should avoid claiming that its diagnostic space is a fractal. The correct register is:

$$\boxed{\text{recursive triadic topology first;}}$$

$$\boxed{\text{fractal-like realization only after scale/readout is declared.}}$$

6G. Count-calibrated threshold face

The ordinal score chart is a declared calibrator. But where possible, the ordinal levels should be tied to countable readouts.

This section adds a count-calibrated face:

$$\Gamma_{\text{count}}$$

as a refinement of the coarse diagnostic granularity Γ_4.

Definition 6G.1 — Raw count readouts

For a declared target regime, define:

Count	Meaning
c_ρ	number of independent active target-face / target-class columns

Count	Meaning
c_b	number of distinct target-presentation classes activated
c_s	maximum recursive role-opening depth
c_n	number of no-free-transfer / type-promotion jumps
c_h	number or severity level of observer/readout/evaluator/calibrator burdens
c_κ	compatibility coverage score

The compatibility coverage score is interpreted separately:

$$c_\kappa = 0$$

for a strong compatibility operator,

$$c_\kappa = 1$$

for a partial / regime-limited operator, and:

$$c_\kappa = 2$$

for absent or contested terminal compatibility.

Definition 6G.2 — Default count-to-ordinal thresholds

The default count-calibrated face is:

Coordinate	L	M	H	VH
ρ_{face} via c_ρ	$0-1$	$2-3$	$4-5$	≥ 6
b_{targ} via c_b	$0-1$	$2-3$	$4-5$	≥ 6
s_{sub} via c_s	0	1	2	≥ 3
n_{NFT} via c_n	$0-1$	2	$3-4$	≥ 5
h_{obs} via c_h	$0-1$	2	$3-4$	≥ 5

These thresholds are not primitive. They are a default count face of:

$$\mathfrak{K}_{\text{diag}}^{\Gamma_{\text{count}}}.$$

Definition 6G.3 — Compatibility coverage

Compatibility is scored by coverage rather than raw difficulty.

Symbol	Count value	Meaning
S	$c_\kappa = 0$	one strong declared operator covers the target regime
P	$c_\kappa = 1$	operator covers some but not all target faces
N	$c_\kappa = 2$	no single operator, or operator contested/absent

Theorem 6G.4 — Count-Threshold Face Admissibility

If a target regime has finite declared target columns, finite declared branch classes, finite substitution depth, finite no-free-transfer jumps, and finite observer/readout burden, then the count-calibrated face Γ_{count} is an admissible refinement of the four-level ordinal diagnostic chart.

Proof

Each diagnostic coordinate is derived from a finite declared target-demand problem. Counting active face columns, branch classes, recursive role openings, type-promotion jumps, and observer/readout burdens gives finite raw readouts. Definition 6G.2 maps those raw counts monotonically into the ordered levels $L < M < H < VH$. Therefore Γ_{count} is a finite ordered quotient/readout of the diagnostic complexity presentation. By the Granularity-Indexed Scoring Calibrator Theorem, it defines an admissible diagnostic calibrator. □

Corollary 6G.5 — Qualitative scoring is permitted only when counts are unavailable

If the raw count readout is available, it should be used. If the target regime is too underdeclared to count, the row should be marked conditional, underdeclared, diagnostic-pending, or qualitative.

Corollary 6G.6 — Scoring disagreement localizes the dispute

If two readers disagree about a row, they should identify which raw count differs:

$$c_\rho,\ c_b,\ c_s,\ c_n,\ c_h,\ \text{or } c_\kappa.$$

This makes disagreement local and repeatable rather than impressionistic.

Worked count example — Maxwellian electromagnetism

For the declared classical EM unification target:

$$c_\rho = 4, \quad c_b = 4, \quad c_s = 1, \quad c_n = 2, \quad c_h = 2, \quad c_\kappa = 0.$$

Thus:

$$(\rho, b, s, n, \kappa, h) = (H, H, M, M, S, M).$$

The demand mass is:

$$Q = 2 + 2 + 1 + 1 + 1 = 7,$$

and S gives $\kappa^- = 0$. Hence the row is finite unification atlas.

Worked count example — Observer-slice AC / continuum node

For the declared target:

observer-slice AC presentation calibrated to multiple stronger targets,

the active demands include order, interval, topology, charting, carrier coherence, dimension, product policy, theorem applicability, count, density, anchoring, metric scale, dynamics, and empirical readout.

A conservative count is:

$$c_\rho \geq 8, \quad c_b \geq 6, \quad c_s = 2, \quad c_n \geq 5, \quad c_h \geq 4, \quad c_\kappa = 2.$$

Thus:

$$(\rho, b, s, n, \kappa, h) = (VH, VH, H, VH, N, H/VH).$$

The table records $h = H$ rather than VH to avoid overclaiming empirical-readout burden at this stage. A refined empirical target may legitimately update this to VH.

6H. Reproducibility scoring sheet

The table is reproducible only if a reader can reconstruct each row from declared scoring conventions.

The diagnostic remains qualitative, but qualitative does not mean arbitrary. This section gives the canonical ordinal scoring convention used by the table.

6H.1 Ordinal values

For the five demand coordinates:

$$\rho_{\text{face}}, \quad b_{\text{targ}}, \quad s_{\text{sub}}, \quad n_{\text{NFT}}, \quad h_{\text{obs}},$$

use:

Symbol	Value	Meaning
L	0	low / one narrow target family
M	1	moderate / several linked demands
H	2	high / many independent demands
VH	3	very high / many independent demands plus deep observer or target branching

For compatibility operator availability:

Symbol	Compatibility status	Obstruction value κ^-
S	strong declared compatibility operator	0
P	partial or regime-limited compatibility operator	1
N	absent, contested, or no single operator	2

Here κ^- measures absence of terminal compatibility, not compatibility itself.

6H.2 Demand mass

Define the **demand mass**:

$$Q(X, D) = \rho_{\text{face}} + b_{\text{targ}} + s_{\text{sub}} + n_{\text{NFT}} + h_{\text{obs}}.$$

With the ordinal convention:

Band	Range	Interpretation
low	$0 \leq Q \leq 3$	bounded target likely
moderate	$4 \leq Q \leq 6$	bounded or finite bridge depending on κ
high	$7 \leq Q \leq 10$	finite unification or mixed branch likely
very high	$Q \geq 11$	critical or underdetermined likely unless strong κ collapses the target

6H.3 Classification rule tree

Given:

$$(Q, \kappa^{-}, \Delta_{\text{diag}}, m_C, \mathcal{D}_{\text{post}}),$$

use the following rule tree.

Rule 1 — Underdeclared first

If the source, target regime, observer slice, or compatibility operator is insufficiently declared, return:

$$\boxed{\text{insufficiently declared}}$$

or:

$$\boxed{\text{diagnostic pending.}}$$

Rule 2 — Bounded bridge

If $Q \leq 6$, $\kappa^{-} = 0$, and no post-diagnostic residual family persists, classify as:

$$\boxed{\text{bounded bridge.}}$$

Rule 3 — Finite unification atlas

If $Q \geq 7$, $\kappa^{-} = 0$, and the compatibility operator collapses the activated target branches into a finite law/system, classify as:

$$\boxed{\text{finite unification atlas.}}$$

Rule 4 — Conditional / mixed finite atlas

If $Q \geq 7$, $\kappa^{-} = 1$, and the compatibility operator closes some but not all activated target branches, classify as:

$$\boxed{\text{conditional / mixed finite atlas.}}$$

Rule 5 — Critical relation locus

If $Q \geq 9$, $\kappa^{-} \geq 1$, and the post-diagnostic pass confirms persistent residuals that generate stable bridge families, classify as:

$$\boxed{\text{critical relation locus.}}$$

If $Q \geq 11$, $\kappa^{-} = 2$, and $\mathcal{D}_{\text{post}}$ is incomplete, return:

$$\boxed{\text{critical / underdetermined}}$$

rather than admitted criticality.

Rule 6 — Borderline

If the best two class scores are within the declared margin:

$$m_C \leq \eta_{\text{class}},$$

return:

$$\boxed{\text{borderline}}$$

or:

$$\boxed{\text{mixed.}}$$

6H.4 Row-reconstruction card

A reader can reconstruct any row using this card:

Field	Reader fills
Case/domain	What is being considered?
Declared target regime	What exact target distinctions are being tested?
Active faces	Which $\mathsf{Z}, \mathsf{R}, \mathsf{O}$ demands are active?
Coordinate trace	$(\rho, b, s, n, \kappa, h)$
Demand mass	$Q = \rho + b + s + n + h$
Compatibility obstruction	κ^{-}
Active theorem/no-go subtree	Which programme theorem family fires?
Post-diagnostic status	residual persists / closes / blocks / pending
Classification	bounded / finite / critical / mixed / underdeclared
Challenge point	Which coordinate or target declaration would change the row?

6H.5 Worked reconstruction: Maxwellian electromagnetism

Row 5 has:

$$(\rho, b, s, n, \kappa, h) = (H, H, M, M, S, M).$$

Convert:

$$(H, H, M, M, M) = (2, 2, 1, 1, 1)$$

for the five demand coordinates, so:

$$Q = 2 + 2 + 1 + 1 + 1 = 7.$$

The compatibility obstruction is:

$$\kappa^- = 0$$

because Maxwell equations provide a strong compatibility law for the declared classical EM target.

By Rule 3:

$$Q \geq 7, \quad \kappa^- = 0 \Rightarrow \text{finite unification atlas.}$$

6H.6 Worked reconstruction: observer-slice AC node

Row 16 has:

$$(\rho, b, s, n, \kappa, h) = (VH, VH, H, VH, N, H).$$

Convert:

$$(VH, VH, H, VH, H) = (3, 3, 2, 3, 2)$$

for the five demand coordinates, so:

$$Q = 3 + 3 + 2 + 3 + 2 = 13.$$

The compatibility obstruction is:

$$\kappa^- = 2.$$

The post-diagnostic pass confirms persistent residuals generating stable bridge families:

$$D_{\text{hard}} = (D_{\mathcal{C}}^{\text{flat/global}}, D_{\text{car}}, D_d^{\text{wit}}, D_{\text{prod}}, D_{\text{thm}}, D_{\text{count}}, D_{\text{dens}}, D_{\text{anchor}}, D_{\text{metric}}, D_{\text{dyn}}, D_{\text{emp}}).$$

By Rule 5:

$$Q \geq 9, \quad \kappa^- \geq 1, \quad \text{persistent residuals confirmed} \Rightarrow \text{critical relation locus.}$$

6H.7 Reproducibility limit

The scoring sheet makes the table reproducible, not infallible.

A reader can challenge a row by changing:

1. the declared target regime;
2. a coordinate score;
3. the claimed compatibility operator;
4. the active theorem/no-go subtree;
5. the post-diagnostic residual status.

If the challenge changes Q, κ^-, or post-diagnostic status, the classification may change. That is intended.

6I. Worked diagnostic-update / failure case

The diagnostic must be able to revise itself. Otherwise it would be a convenient classifier rather than a falsifiable method.

6I.1 Apparent criticality before compatibility discovery

Consider a pre-unification electromagnetic target regime:

$$D_{\text{preEM}} = \{\text{electric effects}, \text{magnetic effects}, \text{induction}, \text{wave propagation}, \text{source constraints}\}.$$

Before a compatibility operator is supplied, the diagnostic may read:

$$(\rho, b, s, n, \kappa, h) = (H, H, M, H, N, M).$$

This would support a provisional status:

$$\boxed{\text{candidate critical / underdetermined.}}$$

6I.2 Compatibility operator discovery

The declaration of Maxwellian field equations supplies a compatibility operator:

$$\kappa^{\text{EM}}_{\text{compat}}.$$

The target then updates to:

$$(\rho, b, s, n, \kappa, h) = (H, H, M, M, S, M).$$

The class changes to:

$$\boxed{\text{finite unification atlas.}}$$

6I.3 Why this is not inconsistency

The pre-unification classification was conditional on:

$$\kappa = N.$$

The later classification is conditional on:

$$\kappa = S.$$

The diagnostic did not fail by changing. It updated the coordinate whose value changed. This is precisely the role of:

$$\text{Update}_{\text{diag}}.$$

Corollary 6I.4 — Criticality is falsified by terminal compatibility

A critical-locus classification is falsified or downgraded if a declared compatibility operator is found that collapses all active target faces to the declared tolerance.

Thus:

$$\boxed{\text{critical} \quad \not\equiv \quad \text{permanently unsolved.}}$$

It means:

$$\boxed{\text{no adequate compatibility operator is currently declared for the target regime.}}$$

7. Replicable procedure

Because the coordinates are derived from bridge-audit machinery, a researcher applying the diagnostic should:

1. declare source X;
2. declare target distinction family D'_Y;
3. fill the target-demand matrix $\mathsf{T}(X, D'_Y)$;
4. compute/estimate:
$$\mathcal{D}_{\text{pre}} = (\rho_{\text{face}}, b_{\text{targ}}, s_{\text{sub}}, n_{\text{NFT}}, \kappa_{\text{compat}}, h_{\text{obs}});$$
5. classify as bounded, finite-unification, candidate-critical, or underdeclared;
6. run relation-first and diagnostic audit;
7. compute:
$$\mathcal{D}_{\text{post}} = (r_{\text{persist}}, g_{\text{bridge}}, a_{\text{diag}});$$
8. compute Δ_{diag} and class margins;
9. admit the classification only if the diagnostic error and margin conditions pass;
10. otherwise return underdeclared, mixed, borderline, conditional, blocked, or diagnostic-pending status.

This makes the diagnostic repeatable rather than retrospective.

---

8. Claims and non-claims

Claims

1. The diagnostic coordinates are derived from relation-first bridge-audit machinery rather than imposed externally.
2. “Diagnostic audit” means the mathematical relation-first checking procedure defined in this paper, not an internal workflow note.
3. The paper has a canonical programme dependency spine linking each diagnostic component to existing programme paper/theorem families.
4. The paper’s components form a single dependency chain from relation-first triadic declaration to charted readout, recursive calibration, scoring, and table application.
5. The diagnostic is a readout of target-face demand, target-presentation valence, substitution depth, no-free-transfer count, compatibility-operator availability, and observer-slice hardness.
6. The diagnostic is recursively updated through programme theorem/no-go activation sourced in the paper stack.

7. Every named diagnostic residual must activate the relevant theorem/no-go subtree before classification can be admitted.
8. The critical-locus action protocol is derived from the critical diagnostic state, not appended as a heuristic.
9. The critical action subspace is generated by atlas-action generators and compatibility-search generators.
10. A critical-locus classification has constructive content: it triggers the two-track protocol $\mathrm{AtlasMap} + \mathrm{CompatSearch}$.
11. Critical-locus progress is measured by atlas articulation, admitted partial closures, localized residuals, and compatibility-operator search.
12. The 36-case diagnostic landscape table is a compact diagnostic trace table, not a full paper-construction audit.
13. The diagnostic landscape now includes row-level action implications.
14. A reader can reproduce or challenge a row by using the declared target regime, coordinate trace, active theorem/no-go subtree, class, status, residual/guardrail, and action implication.
15. The diagnostic score sheet has a programme-native general form as a granularity-indexed charted calibrator family.
16. The ordinal L, M, H, VH score sheet is a declared coarse diagnostic chart, not a unique theorem forced by RFOP.
17. The count-calibrated threshold face Γ_{count} is an admissible refinement where raw count readouts are available.
18. Diagnostic score charts are quotients/readouts of a recursive triadic topology generated by face demand, role substitution, landed coupling, residuals, traces, and readouts.
19. A fractal-like interpretation of this topology is admissible only after scale, self-similarity, and measure/readout data are declared.
20. The reproducibility scoring sheet gives a declared ordinal convention, demand mass, compatibility obstruction, and rule tree for reconstructing table rows.
21. Criticality can be predicted from target-demand structure before solving the bridge when the diagnostic has sufficient declaration and margin.
22. Persistent residualization confirms criticality only after diagnostic audit and recursive update.
23. The diagnostic distinguishes bounded bridges, finite unification atlases, critical relation loci, accidental underdeclaration, and mixed/borderline/underdetermined cases.
24. The observer-slice AC/continuum node is predicted and confirmed as critical under this diagnostic.
25. Maxwellian and Standard Model-style cases are better read as finite unification atlases, not low-valence bounded bridges, under their declared target regimes.
26. Newtonian mechanics is bounded inside its declared target regime.
27. The appendices provide standalone dependency summaries and representative external literature anchors.

Non-claims

1. The diagnostic does not always classify.
2. A critical-locus classification does not prove permanent unsolvability.
3. A critical-locus classification does not license stopping compatibility-operator search.
4. The two-track protocol is not the whole action space; it is the minimal generating response for critical loci.
5. The 36-case table is not a full paper-construction audit.
6. The recursive diagnostic does not require applying every theorem to every case.
7. Borderline, mixed, insufficiently declared, diagnostic-pending, and granularity-sensitive statuses are valid outputs.
8. The schematic landscape table is not a full historical or technical treatment of each listed case.
9. The row-level action implications are methodological defaults, not completed research programmes.
10. The scoring sheet makes the table reproducible, not infallible.
11. The four-level scoring sheet is not the only possible diagnostic calibrator.
12. Finer or coarser score charts may be admissible if declared and monotone.
13. The count-calibrated thresholds are a default diagnostic face, not an immutable axiom.
14. The paper does not claim that recursive triadic topology is literally fractal.

15. Fractal-like claims require additional scale/readout/self-similarity data.
16. The external references are representative anchors, not load-bearing proofs of the diagnostic.
17. This paper does not prove physical theories from RFOP.
18. This paper does not claim the AC/continuum node is physically fundamental.
19. This paper does not claim all hard problems are critical loci.
20. This paper does not claim residuals are automatically meaningful.
21. This paper does not remove the need to solve individual target-bridge residuals.

Appendix A. Standalone dependency summaries

This appendix gives short summaries for readers who have not read the full programme stack.

A.1 Theory of Organization

Theory of Organization supplies the relation-first base layer: organization is not primitive objecthood but proper non-boundary structure induced by relation, question, answer, and boundary/readout roles. It introduces the discipline that meaningful organization lives away from lower-boundary collapse and upper-boundary totalization.

A.2 Charted Organization Theory

Charted Organization Theory makes organization observer-relative without making the observer a conscious subject. A chart is a relation-map internal to a carrier. An observer is an admissible positioned chart. Information is chart-visible delineation. Objective content is nonvacuous invariance across declared chart transitions. This is the source of the diagnostic's readout/chart discipline.

A.3 Recursive Chart Learning Theory

Recursive Chart Learning Theory defines learning as admitted path tracing through a chart-visible branch field. It distinguishes transition attempts from landed/admitted steps, evaluation from improvement, calibration from path type, and recursion from mere finite sequence. This is the source of diagnostic update, branch fields, and residual-to-learning discipline.

A.4 Recursive Bridge Calibration

Recursive Bridge Calibration is the programme's method for turning a compressed claim:

$$X \Rightarrow Y$$

into:

$$X - \mathfrak{B} - Y.$$

It requires source, target distinction, bridge relation, boundary/tolerance, observer/readout, evaluator, admission rule, and update rule. It is the source of calibrator-relative classification and branch selection.

A.5 Triadic Closure and Recursive Residual Theory

Triadic Closure supplies the primitive role set:

$$\mathcal{P} = \{\mathsf{Z}, \mathsf{R}, \mathsf{O}\}.$$

Stable declared relation requires boundary, mediation, and observer/readout. Singleton and dyadic subfamilies are closure-incomplete. This is the source of face-demand rank and the residual-recursive basis of the diagnostic.

A.6 Triadic Substitution, Coupling, and Closure Atlases

Triadic Substitution says a nontrivial role substitute must itself be triadically declared. Coupling between triads requires a landed closure-complete interface. Closure atlases organize local triads, couplings, residuals, admissible paths, and updates. This is the source of substitution depth, recursive triadic topology, and the score-chart quotient structure.

A.7 No Undeclared Relation

No Undeclared Relation blocks the use of invisible arrows. A bridge claim must declare the relation, target distinction, boundary, observer/readout, and admission conditions. This is the source of the diagnostic's refusal to classify underdeclared bridge problems.

A.8 No Undeclared Boundary

No Undeclared Boundary says no claim of exactness, approximation, inside/outside, admissibility, or failure is meaningful without a declared boundary, tolerance, support, or target distinction. This is the source of diagnostic error, abstention, and local-exactness discipline.

A.9 No Undeclared Observer

No Undeclared Observer says no visibility-bearing inference is legitimate without a declared observer/readout/evaluator. This is the source of h_{obs}, the scoring calibrator, and the requirement that diagnostic readouts be declared.

A.10 No-Free-Transfer / structure-type no-promotion

No-Free-Transfer says that accessibility for one target type does not imply accessibility for another. Order does not imply dimension. Counting does not imply continuum volume. Metric does not imply dynamics. This is the source of n_{NFT}.

A.11 Bridge Accessibility and Target-Relative Inaccessibility

Bridge Accessibility states that bridgedness is admitted target accessibility, and unbridgedness is target-relative inaccessibility. This is the semantic basis of b_{targ} and of the diagnostic's target-regime relativity.

A.12 Target-Presentation Calibration Atlas

The Target-Presentation Calibration Atlas fixes a source presentation and compares it to multiple declared target presentations under one residual language. This is the source of the diagnostic landscape structure and of the table's target-regime discipline.

A.13 Critical Relation Loci and Bridge-Valence

Critical Relation Loci introduced the idea that persistent residualization can indicate a high bridge-valence relation-presentation inside an observer slice. The present paper strengthens that idea into a predictive diagnostic with error, abstention, recursion, scoring, and falsification checks.

Appendix B. External calibration literature for the landscape table

The 36-case table is schematic and target-regime-relative. The following references are representative external anchors for selected cases. They are not load-bearing proofs of the diagnostic.

B.1 Classical and field unification cases

- Maxwell, J. C. (1865). "A Dynamical Theory of the Electromagnetic Field." *Philosophical Transactions of the Royal Society of London*, 155, 459–512.
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B.5 Methodology cases

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B.6 Relation to Lakatos

The bridge-valence diagnostic is adjacent to Lakatos's research-programme methodology, but it does not duplicate it. Lakatos distinguishes progressive from degenerating research programmes by theoretical and empirical progress. The present diagnostic distinguishes bounded bridges, finite unification atlases, and critical loci by target-demand structure, compatibility operators, and residual persistence. The connection is methodological: both frameworks ask whether a programme's response to difficulty is progressive or merely protective. The difference is that this paper supplies a relation-first local diagnostic for bridge problems rather than a historical appraisal of whole research programmes.

Appendix C. Version and review-readiness note

This v1.9 draft supersedes earlier draft labels. Earlier version notes in the abstract describe the development path only. The manuscript-level version is v1.9.

The remaining expected editorial work is citation stabilization and compression, not conceptual repair.

9. Final statement

By the cohesion theorem, the diagnostic is not an external method appended to RFOP; it is a charted, recursive, calibrated readout of RFOP bridge machinery.

A critical locus is not a label for difficulty and not a terminal verdict. It is an instruction to run atlas articulation and compatibility-operator search in parallel.

A critical locus is not a label for difficulty.

It is a relation-first diagnostic class, and the diagnostic is programme-derived, recursive, partial, and error-bounded rather than total:

$$\boxed{\text{high face-demand rank} + \text{high bridge-valence} + \text{deep substitution} + \text{many no-free-transfer jumps} + \text{no termina}}$$

$$\boxed{\Rightarrow \text{branching target atlas.}}$$

That is the rigorous version of the intuition that some loci are not finite-answer targets but crossroads of admissible relation-paths inside an observer slice.

End draft v1.12

AI-assistance disclosure

This manuscript was developed with AI-assisted drafting and review support. The author directed the research programme, selected and revised the mathematical and methodological content, and is responsible for all claims and errors.